



BASE24-card[®]

Reference Manual

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Preface

This manual provides an overview and technical description of the BASE24-card release 6.0 product. This product is designed to allow financial institutions the ability to control the issuing and maintenance of plastic cards on the Tandem computer system.

Audience

ACI has prepared this publication for distribution to personnel responsible for the plastic card issuance and control functionality of the BASE24-card product. This publication also provides an overview of the BASE24-card system for those persons interested in gaining a general understanding of the system.

Prerequisites

Readers should be familiar with the other BASE24 products (i.e., BASE24-atm and BASE24-pos) that can be used in conjunction with the BASE24-card product. This knowledge allows readers to better understand functions of the various files and processes described in this manual.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and, at the same time, provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The ***BASE24-from host maintenance Reference Manual*** provides detailed information on the processing of the BASE24-from host maintenance product.
- The ***BASE24 Refresh and Extract Operations Manual*** provides detailed information on the processing of the BASE24 Refresh and Extract processes.

This manual also contains references to the following International Organization for Standardization (ISO) publications:

- The ISO 4909 standard, *Magnetic Stripe Data Content for Track 3*, provides detailed information on the format of Track 3 data.
- The ISO 3166:1993 standard, *Codes for the Representation of Names of Countries*, lists valid ISO country codes.
- The ISO 4217:1995 standard, *Codes for the Representation of Currencies and Funds*, lists valid ISO currency codes.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

Section 1, “Introduction,” provides an introduction to the BASE24-card product, including an overview of the BASE24 application files, control terminals and programs. It also discusses the embossing, encoding, indenting formats and the reporting provided by the BASE24-card product.

Section 2, “Operations,” provides a more detailed look into the BASE24-card operation procedures, including base file maintenance, card management programs and card production examples.

Section 3, “Embossing, Encoding, and Indenting Formats,” contains the formats for the configuration of embossed, encoded, and indented information on the plastic cards.

Section 4, “Card Reporting,” describes the reports that the BASE24-card product provides.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, CT-AE000-01 for the ***BASE24-card Reference Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, 09/2001 for September, 2001), edition, and software release information (for example, R6.0v1 for release 6.0, version 1) identifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

ACI Worldwide Inc.

Section 1

Introduction

The BASE24-card product is a card issue and control system that enables institutions to control the issuing and maintenance of plastic cards on the Tandem computer system. It also provides the capability of maintaining the card database while online transaction authorization occurs.

This section provides an introduction to the BASE24-card product.

Overview

The BASE24-card product provides a flexible plastic card management system for financial institutions and retailers. The BASE24-card product also enables an institution to provide a complete plastic card processing operation concurrent with a BASE24-atm and/or pos transaction authorization system on a single Tandem computer system. The BASE24-card product was designed with the flexibility to accommodate virtually any card management scheme. This flexibility is especially important in today's competitive environment where businesses are providing electronic funds transfer services and are guided by their own electronic funds transfer (EFT) marketing and operational strategies. The BASE24-card product can be implemented for whatever method an institution chooses to use in issuing, controlling, and retiring its ever-expanding base of plastic cards. For example, some institutions use an annual reissue cycle, while others use a two-year cycle. The BASE24-card product can easily accommodate either situation. Furthermore, the system's user selectable options enable an institution to market card-issue services to other institutions, without requiring those institutions to conform to any one standard.

Other BASE24-card features allow institutions the latitude to include on the DataCard embossing machines those features that best meet their individual needs. In addition, PIN mailers and card carriers can be produced in-house on a printer hooked up to the Tandem processor.

Each cardholder account can be configured to produce any combination of a plastic card, a PIN mailer, and a card carrier (only one of these three, any two of these three, or all three). This is beneficial, for example, when a cardholder forgets his or her PIN. A PIN mailer could be generated without the plastic card or the card carrier.

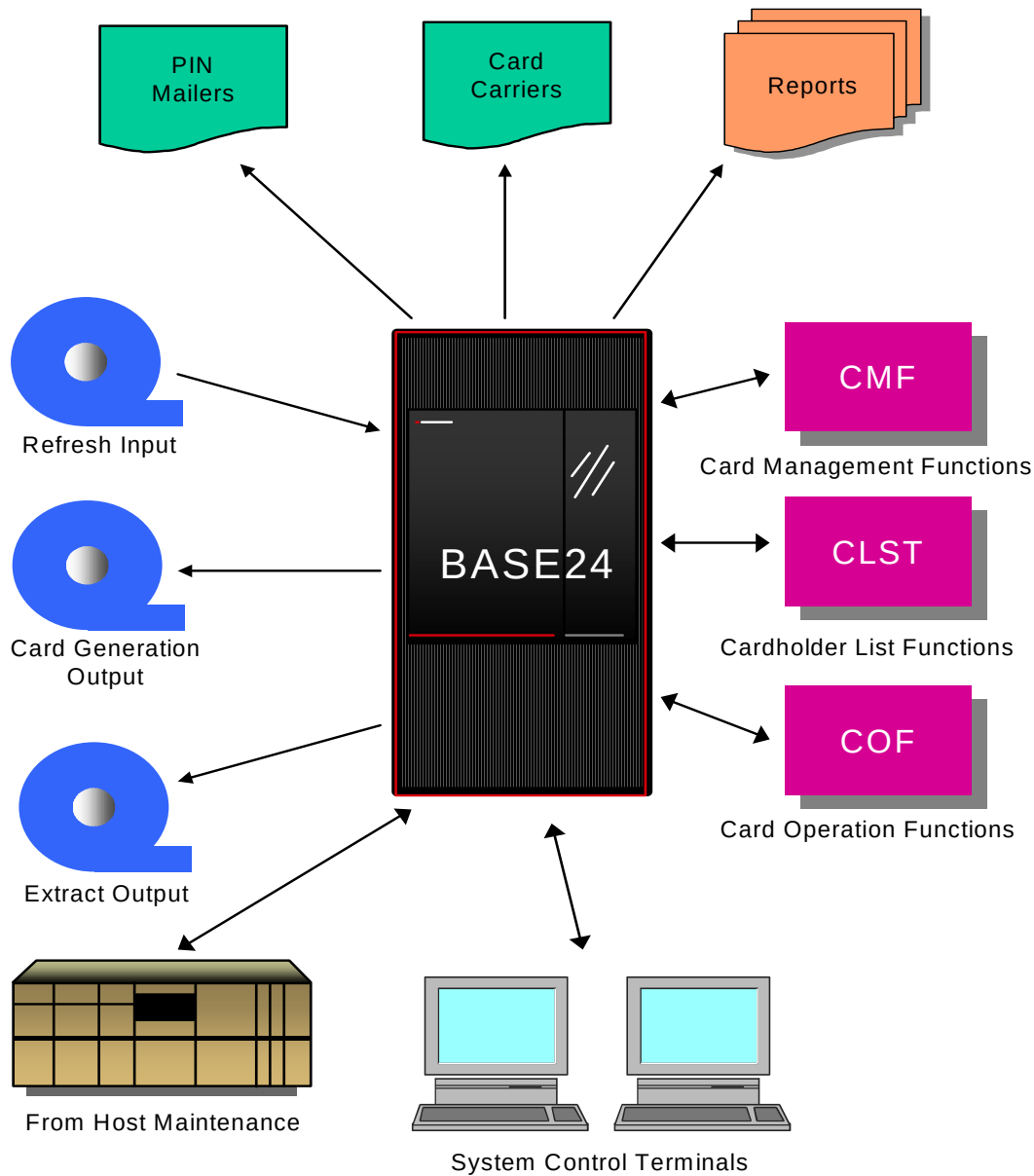
The BASE24-card product is integrated with other BASE24 products into a single system. As an integrated system, all products access a common database whenever possible. The BASE24-card product equips an institution with state-of-the-art technology geared to the EFT and card issuance needs of the present and beyond. The system is designed to evolve and grow, thereby satisfying the ever-changing needs of the institution. New versions of the BASE24-card product will be released periodically to upgrade system performance and to provide support of the latest in card issuance technology.

As a comprehensive card issue and control system, the BASE24-card product enables the Tandem processor to perform many functions. These functions include the following:

- Issue of new plastic cards, card carriers, or PIN mailers
- Force issue of plastic cards, card carriers, or PIN mailers
- Reissue of plastic cards, card carriers, or PIN mailers
- Mass reissue of plastic cards, card carriers, or PIN mailers
- Support of multiple card types per institution
- Automatic generation of possible card account numbers
- Support of user-selected length of issue
- Automatic generation of issue reports
- Automatic generation of embossing, encoding, and indenting tapes
- Automatic generation of production and control reports
- Automatic purging of expired cards
- CRT entry for database maintenance
- Magnetic tape input for database maintenance
- BASE24-from host maintenance access to the database

System Components

The following picture shows the main components and functions of a BASE24-card system. The remainder of this publication examines these components further and describes the basic functionality of each.



Card Management Control Terminals

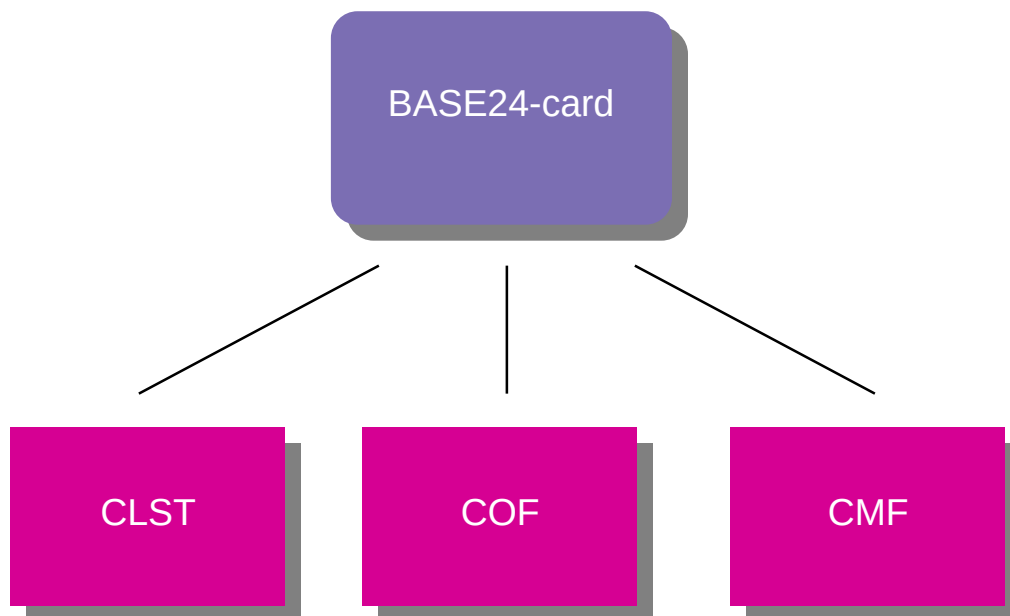
There are three CRT terminal functions, which provide human interface to the card management functions. They are:

- Cardholder List Functions (CLST)
- Card Operator Function (COF)
- Card Management Functions (CMF)

As with other BASE24 system control CRTs, the card management terminals are driven by Tandem's NonStop Pathway software. This enables the use of pre-formatted screens, providing easy operator interface. The card management terminals are also part of BASE24's comprehensive CRT security system, whereby each operator is issued a user name (logon ID) that permits entry into only those file and function screens for which he or she is authorized as a member of a particular institution.

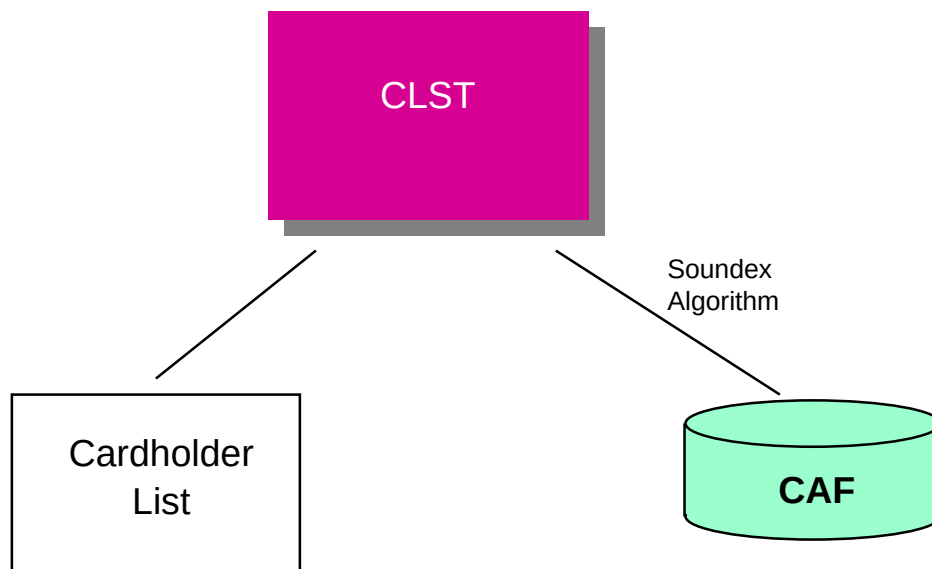
The following pages describe each of the card management terminals.

Detailed information on these terminal functions is provided in section 2. The functions available from card management control terminals are shown in the following illustration.



Cardholder List Functions (CLST)

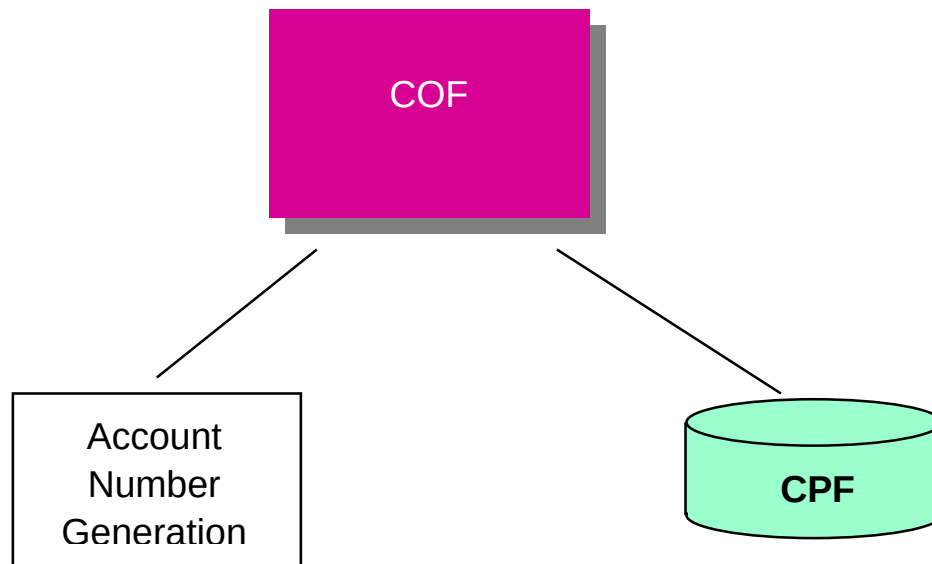
The Cardholder List Functions (CLST) terminal is used to locate a cardholder's account number by entering his or her name. The CLST gives an operator access to the Cardholder Authorization File (CAF) records using only a name. It is designed for situations where an operator is not in possession of an account number but has at least an approximate spelling of a cardholder's name. The CLST server utilizes the Soundex algorithm (which is described in detail in the Soundex Algorithm subsection) to achieve this functionality, which is initiated by pressing a function key from a CLST terminal work screen.



Card Operator Function (COF)

The Card Operator Function (COF) terminal is used to initiate the generation of account numbers. The account number generation function is performed by the SVAGEN server, which is initiated using a function key from a COF terminal work screen. The SVAGEN program can be run at any time and is used to generate a list of possible card account numbers. The COF screen is used to enter the prefix for which the account numbers are to be generated and the number of account numbers to be generated. The SVAGEN program then generates a group of available account numbers that can be assigned to new card accounts. When SVAGEN creates the possible new account numbers, it uses Mod 10 checking to verify that the card numbers are valid. Output from the SVAGEN program can be in the form of a list or, optionally, a set of labels. The LAST ACCOUNT NUMBER field on Cardholder Prefix File (CPF) screen 20 is updated with the last account number generated by the SVAGEN program.

The card operator function is depicted in the following illustration.



Card Management Functions (CMF)

The Card Management Functions (CMF) terminal enables an operator to perform the following functions:

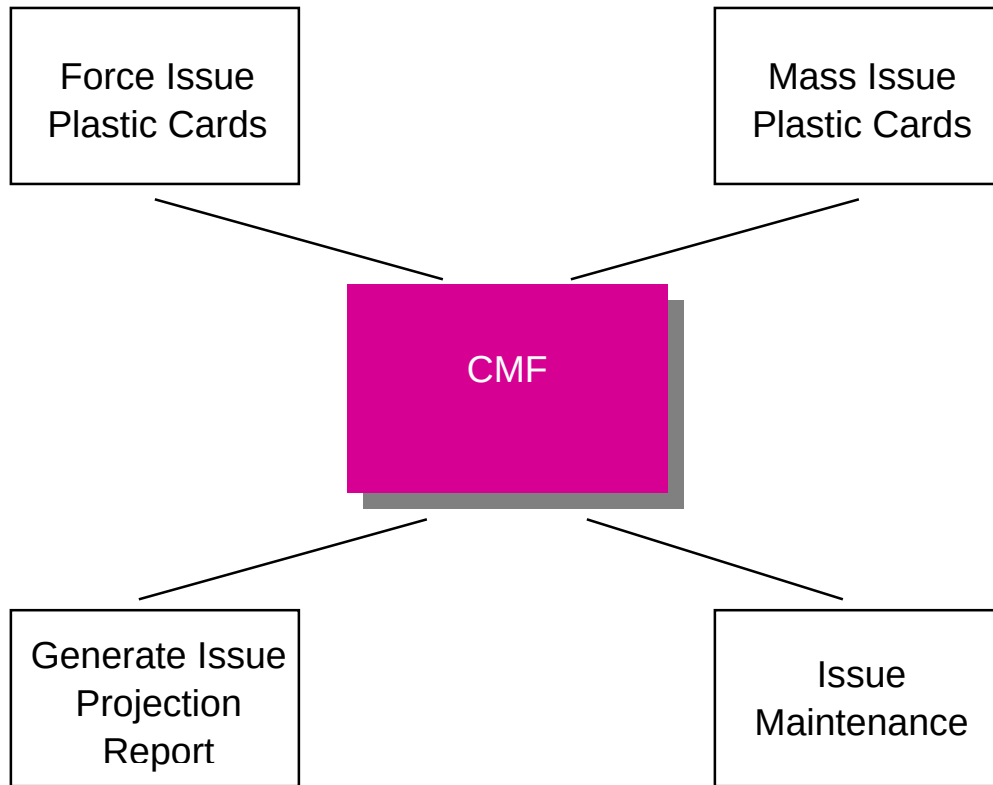
- Generate the Issue Projection Report
- Perform issue maintenance
- Create a plastic-card-generation force or mass issue tape or disk file

The Issue Projection Report contains a month-by-month total of cards that have either been issued since the beginning of the current month or are due for reissue in coming months (the report covers a 12-month period starting with the current month of the current year). The Issue Projection Report can be used to forecast supply needs for card production as well as to evaluate the institution's current card reissue schedule.

The issue maintenance function allows the operator to perform a variety of tasks, all of which involve examining individual card accounts against issue or reissue criteria to determine whether a card should be issued and whether or not PIN mailers and card carriers should be generated. Accounts not meeting established issue/reissue criteria are listed on an exception report allowing institutions to evaluate them and take actions as necessary. Accounts passing the issue/reissue criteria are listed on a report informing institutions which cards will be produced when the plastic card generation process is executed.

The plastic-card-generation force and mass issue functions are performed from a CMF terminal work screen. They utilize the Plastic Card Generation Server (SVPCGEN) program to create either a magnetic tape or a disk file containing the information necessary for a card embossing machine to emboss, encode, and indent plastic cards. The Tandem spooler location or disk file can also, optionally, contain data for PIN mailer and card carrier production. Output can also be directed to an online DataCard embossing machine for real-time card production. The force issue function enables plastic cards to be individually produced in emergency situations such as loss, theft, or damage. The mass issue function enables an institution to reissue cards for its entire card base or for certain card prefixes.

Card management functions are depicted in the following illustration.



Card Management Programs

This subsection describes each of the programs that make up the BASE24-card system. These programs are as follows:

- Account Number Generation Server (SVAGEN)
- Card Management Functions Server (SVCMF)
- Cardholder List Server (SVLST)
- Plastic Card Generation Server (SVPCGEN)

Each of these programs is briefly described in this subsection. For more detailed information on these programs, refer to section 2.

In addition to the card management programs, the BASE24-card product impacts existing BASE24 products and processes because of the card management data that is added to the Cardholder Authorization File (CAF) and Card Prefix File (CPF) as extensions to support the BASE24-card functions. The BASE24 processes impacted by the BASE24-card product include the following:

- BASE24-from host maintenance
- Enscribe Refresh
- Super Extract

Processing and operation information for the above processes can be found in the following manuals: ***BASE24-from host maintenance Reference Manual*** and ***BASE24 Refresh and Extract Operations Manual***, respectively.

Please refer to the CAF and CPF Data Definition Language (DDL) files to review the record layouts of card management data fields.

Account Number Generation (SVAGEN)

The Account Number Generation (SVAGEN) program generates a report of possible new account numbers for a particular prefix. The primary purpose of SVAGEN is to derive account numbers that the institution can assign to new accounts being opened. When SVAGEN creates the possible new account numbers, it uses Mod 10 checking to verify that the card numbers are valid. As an option, the account numbers, which SVAGEN generates, can be printed as a listing or individually on labels.

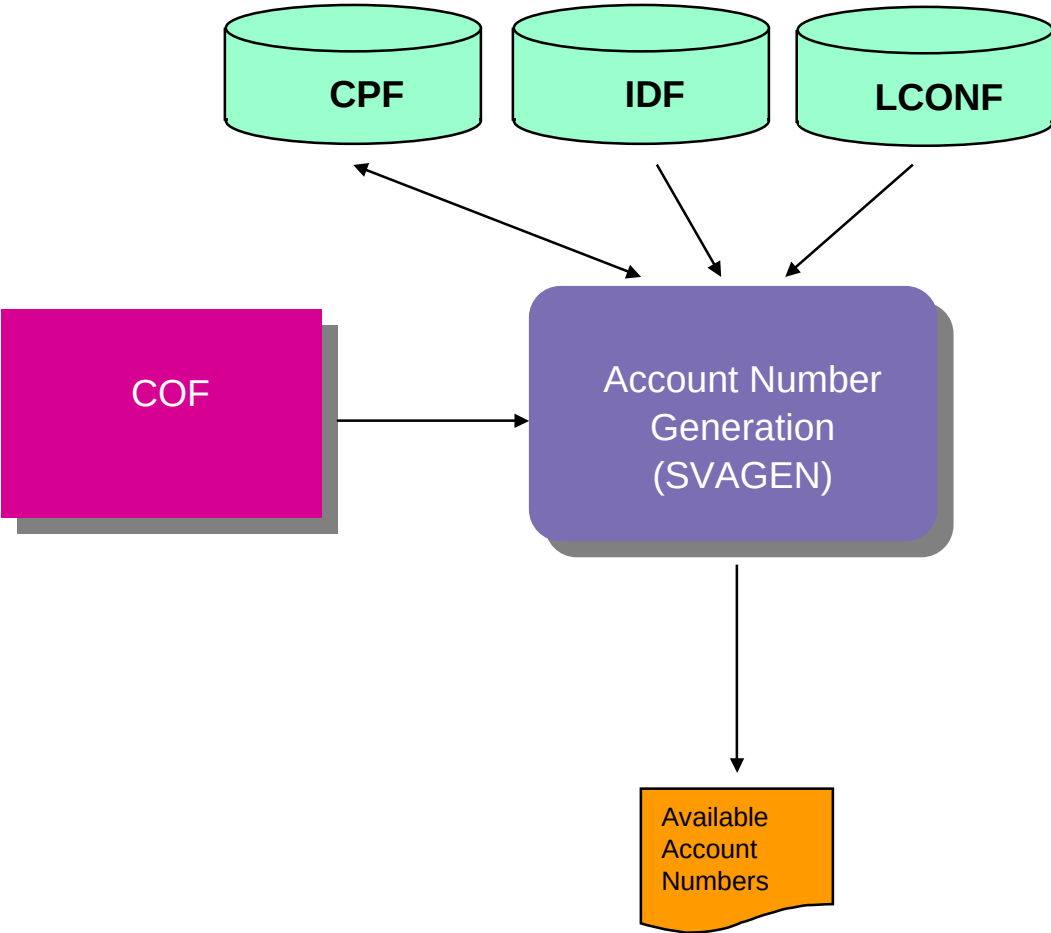
Account number generation is initiated using the Card Operator Function (COF) terminal. To do so, an operator must enter the prefix for which the account numbers are to be generated, the number of account numbers to be generated, and whether the output is to be produced on labels or as a listing.

If the institution chooses to have the output printed on labels, SVAGEN prints one account number per each individual label. When SVAGEN is directed to print the account numbers as a listing, additional headings and space are provided so that an employee can fill in additional information as each number is assigned. This information includes the name of the primary cardholder to which the account number is assigned, the name of the institution employee who assigned the account number, and the date the number was assigned. More information on the Available Account Numbers Report can be found in section 4.

To generate the list of new account numbers, SVAGEN accesses the Card Prefix File (CPF) and the Institution Definition File (IDF). The CPF is read to validate the prefix for which the account numbers are to be generated. In addition, the LAST ACCOUNT NUMBER field on CPF screen 20 is checked and used as the starting point for number generation. The IDF is used to validate institutional parameters for the institution associated with the prefix.

When all the requested account numbers have been generated, the LAST ACCOUNT NUMBER field on CPF screen 20 is updated to reflect the last account number generated.

The Account Number Generation (SVAGEN) program is depicted in the following illustration.



Card Management Functions Server (SVCMF)

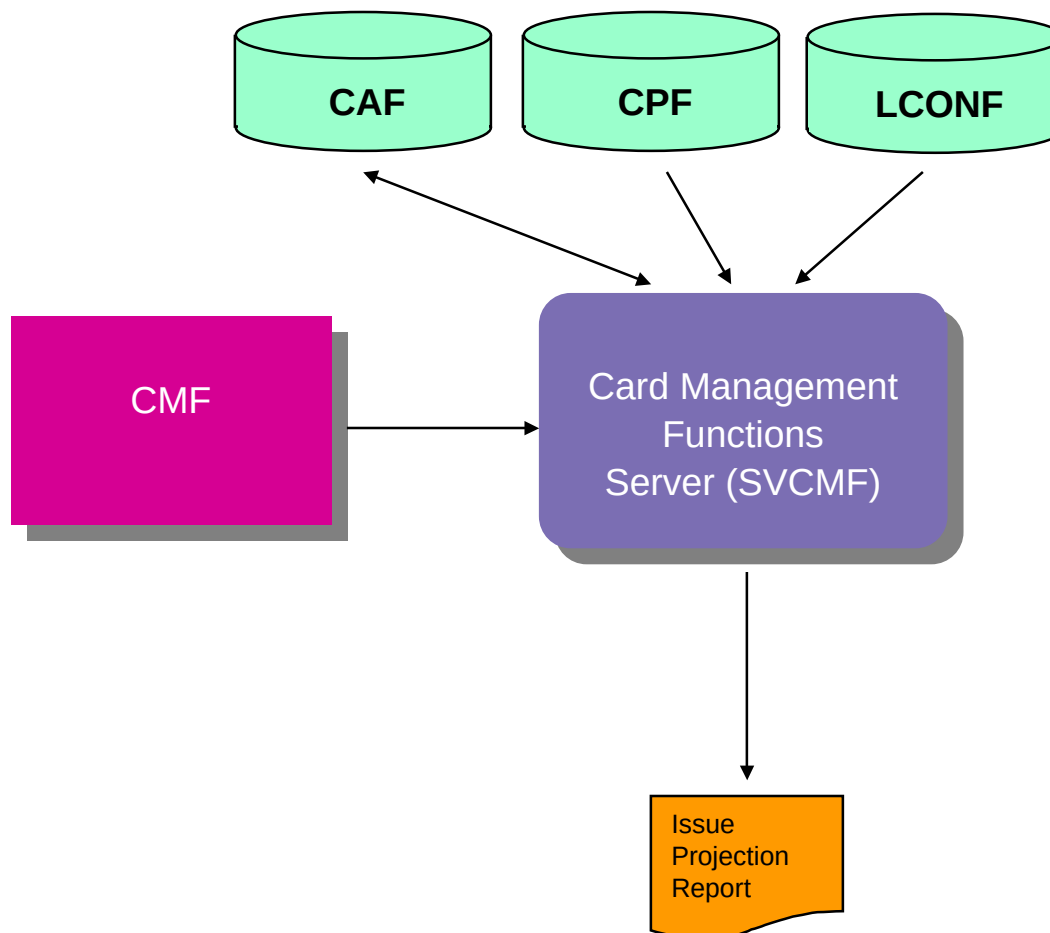
The Card Management Functions Server (SVCMF) program performs two main functions. The program generates the Issue Projection Report and performs issue maintenance. Each of these functions is described below.

Issue Projection Report

The SVCMF program generates a hard copy report containing a month-by-month total of cards that have either been issued since the beginning of the month or are due for reissue in coming months (the report covers a 12-month period starting with the current month of the current year).

The Issue Projection Report can be used to forecast supply needs for card production as well as to evaluate the institution's current card reissue schedule. In the latter case, the report can assist institutions in achieving a uniform card production schedule, whereby relatively the same number of cards is reissued each month.

The issue projection function is initiated using a function key on the Card Management Function (CMF) screen. The SVCMF program accesses data in the Card Prefix File (CPF) and the Cardholder Authorization File (CAF) to generate the Issue Projection Report as shown in the following illustration.



Issue Maintenance

The issue maintenance function includes a combination of functions which enable the institution to evaluate accounts for which plastic cards are being issued. The SVCMF program evaluates these accounts against a set of institution-defined parameters and either allows the generation of the card or reports the account as an exception for possible action by the institution.

For issue maintenance, the SVCMF server performs the following checks:

- Verifies that the card segment is present in the CPF
- Ensures that if the PIN CHECK TYPE field on CPF screen 2 is set to a value of 04 (Visa PVV), the PIN LENGTH field on CPF screen 20 must be set to a value of 4 and the PIN GENERATION TYPE field on CPF screen 20 must be set to a value of R (Random PIN Generation)

If any of these checks fail, the SVCMF server will stop processing the request and the operator will receive a message on the CMF requester screen. After altering the CPF record or records, the operator can reissue the command.

The issue maintenance function includes other maintenance tasks dealing with the purging of expired accounts and the reporting of “out of the ordinary” accounts about to enter their production cycles.

In a number of cases, the SVCMF is the program responsible for creating the GEN-KEY field values (the GEN-KEY fields are alternate keys in the CAF for specific use by the BASE24-card product) for accounts being issued cards or PIN mailers. (The SVPCGEN program uses these GEN-KEY fields to actually create the embossing/encoding/indenting, card carrier, and PIN mailer records).

When the issue maintenance function is run, it performs the indicated checks on accounts for the following conditions:

- Accounts that have expired.

If a card account is expired and the CPF EXPIRATION CYCLE field is set to zero, the account is listed on the Exception Report by the issue maintenance function. If the institution has set the CPF EXPIRATION CYCLE field to a number of days and it is past that number of days, the account is listed on the Exception Report and also purged from the CAF.

- Accounts in the production cycle.

The production cycle is defined by the institution as the period of time prior to expiration during which an account is eligible for reissue of a new card.

If a card account is in its production cycle, the issue maintenance function checks to determine whether the account meets the institution's criteria for reissue. If the account meets the institution's criteria for reissue, its GEN-KEY is created and written to the CAF and the account is listed on the Issue Maintenance Report. The CAF issue status is updated to 07 (Due for Reissue). When the Plastic Card Generation Server is run against this record, a new plastic card is required with a new expiration date. (The expiration date of the CAF is increased by the value of the LENGTH OF ISSUE field on CPF screen 20.) The PLASTIC/CARD CARRIER OPTIONS field on CAF screen 20 must be set to produce a plastic card for the record. If the account fails to meet the criteria necessary for it to be reissued, the GEN-KEY is not created, and the account is listed on the Exception Report.

- Accounts in the reissue cycle.

The reissue cycle is defined by the institution as the period of time prior to the production cycle when accounts are evaluated against the institution's reissue criteria to isolate those accounts that cannot be reissued unless action is taken.

If a card account is in its reissue cycle, the issue maintenance function checks to determine whether the account meets the institution's criteria for reissue. If the account fails to meet the institution's criteria for reissue, it is listed on the Exception Report.

- Newly issued or force-issued accounts.

If a card account is being newly issued or force-issued, the issue maintenance function checks to determine whether the account meets the institution's criteria for card issue. If the account meets the institution's criteria for card issue, the GEN-KEY is created and written to the CAF and it is listed on the Issue Maintenance Report. If the account fails to meet the institution's criteria for card issue, the GEN-KEY is not created, and the account is listed on the Exception Report.

Newly issued and force-issued accounts are identified by a value of 01 (New Issue) or 02 (Force Issue) in the ISSUE STATUS field on CAF screen 20. The issue maintenance function checks these accounts against the institution's normal issue criteria to ensure their suitability for receiving cards.

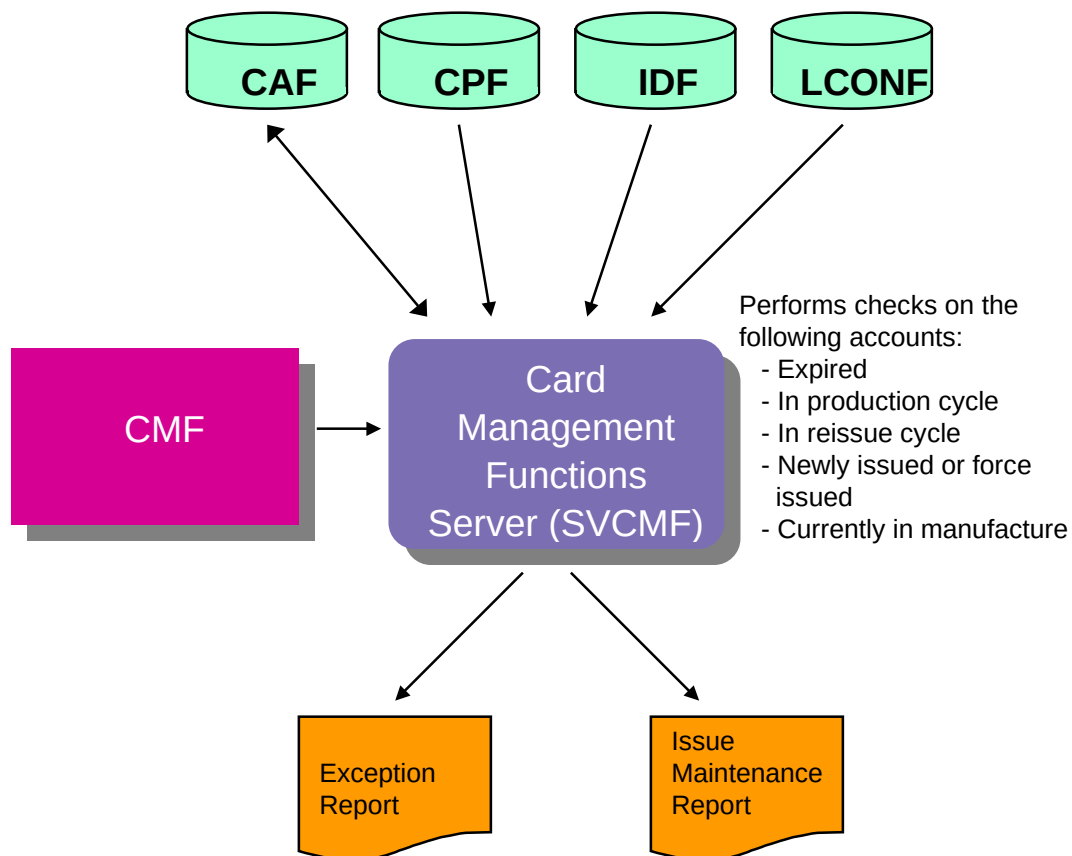
- Accounts currently in manufacture.

If a card account has a value of 11 (New issue in manufacture), 12 (Force issue in manufacture), 13 (Mass issue in manufacture), or 17 (Reissue in manufacture) in the ISSUE STATUS field on CAF screen 20, the issue maintenance function checks to determine if the date of issuance for this card is prior to the current date. If the date the card was issued is prior to the current date, the ISSUE STATUS field is set to a value of 00 (Normal). This would mean the card has been issued and can be evaluated again in the future. If the date the card was issued is not prior to the current date, the account is listed in the Exception Report. This means the plastic is still being produced and will not be reissued again at this time. The date the card was issued is identified in the ISSUE DATE field on CAF screen 20. This date is updated during the plastic card generation processing.

The issue maintenance function accesses the following files:

- Cardholder Authorization File (CAF)
- Card Prefix File (CPF)
- Institution Definition File (IDF)

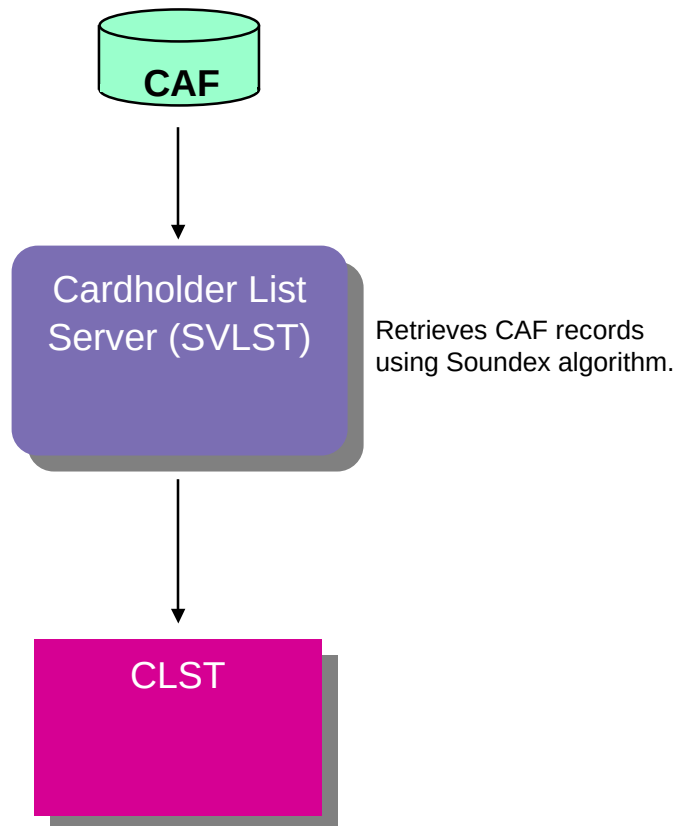
The issue maintenance function is initiated using a function key on the Card Management Function (CMF) screen. The processing performed by the SVCMF during issue maintenance is shown below.



Cardholder List Server (SVLST)

The Cardholder List Server (SVLST) allows operators to list and select cardholder accounts from the Cardholder Authorization File (CAF) for which card maintenance needs to be performed. It is used when an operator is not in possession of an account number but has at least an approximate spelling of a cardholder's name. The server uses the Soundex algorithm to return a grouping of similar names and their corresponding card account numbers and member numbers from which the operator can choose. The algorithm operates on a phonetic degree of similarity.

The cardholder list function is initiated when an operator enters an approximate spelling of the cardholder name on the Cardholder List (CLST) screen and presses a function key. The SVLST program uses the Soundex algorithm to access account data in the Cardholder Authorization File (CAF) and display it on the CLST. The operator can then select one of the account records to be displayed on Cardholder Authorization File (CAF) screen 1. The functions of the Cardholder List Server are shown in the following illustration.



Plastic Card Generation Server (SVPCGEN)

The Plastic Card Generation Server (SVPCGEN) enables an institution to generate plastic cards, PIN mailers, and card carriers. More specifically, SVPCGEN can perform the following functions:

- Generate either an output file (on disk or tape) containing the information necessary for an institution to emboss, encode, and indent plastic cards or send the information to emboss, encode, and indent plastic cards directly to a DataCard embossing machine.
- Produce four-digit PINs with software PIN generation, or with the Thales e-Security (Racal) HSM using ABA Visa PIN generation.
- Produce variable-length random (with a PIN offset) or natural (PIN offset of zeros) PINs with the Thales e-Security (Racal) HSM using IBM DES PIN generation.
- Produce PIN mailers using a software method or with a Thales e-Security (Racal) HSM.
- Generate card carriers on the Tandem processor.
- Create the card verification digits (CVD), which can be embossed or encoded onto the plastic using either a software method or a hardware security device.
- Create the manual card verification digits (CVD), which can be indented onto the plastic using either a software method or a hardware security device.
- Generate a Control Report containing information on the plastic cards created.

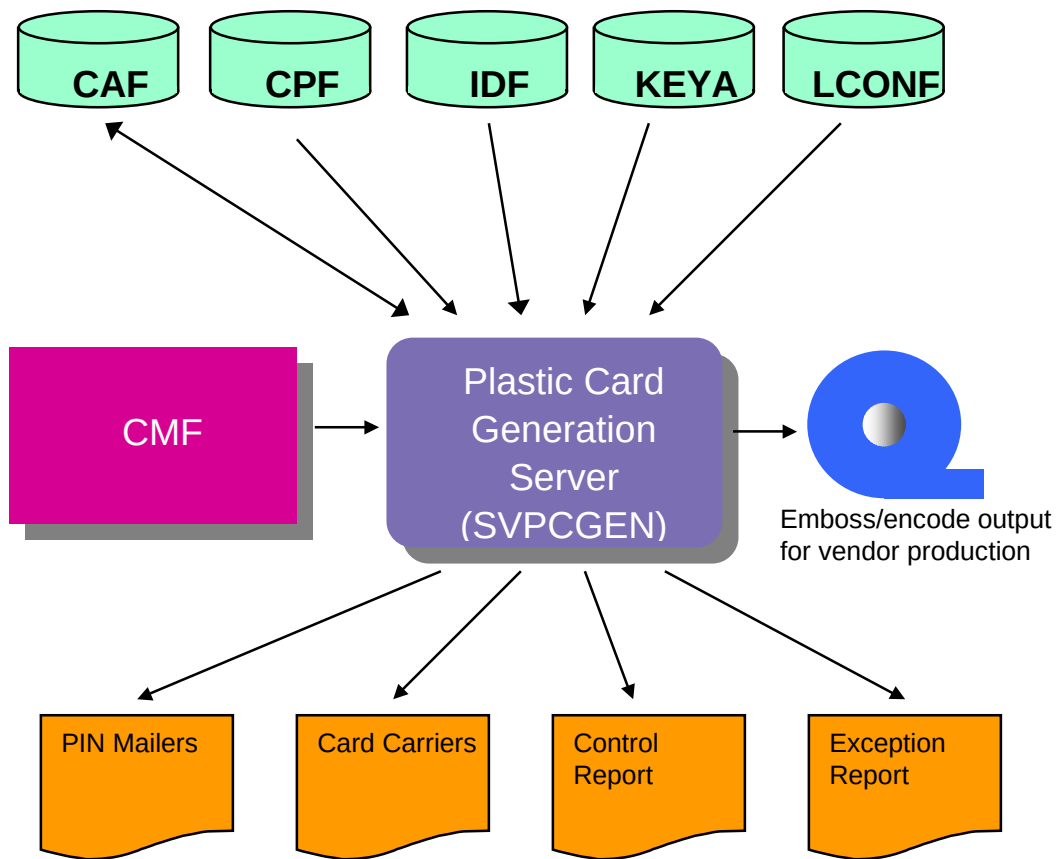
SVPCGEN is initiated using function keys from the Card Management Functions (CMF) screen. It uses the GEN-KEYs created by the issue maintenance function to create the Embossing, Encoding, and Indenting records it places in the output file. The BASE24 standard Open Account Relationship (OAR) functionality is transparent to the BASE24-card product; therefore, plastic cards can be created with up to 16 accounts of the same type associated with the card.

The following files are accessed when the plastic card generation function is executed:

- Cardholder Authorization File (CAF)
- Card Prefix File (CPF)
- Key Authorization File (KEYA)
- Institution Definition File (IDF)

For more information on running the plastic card generation function, refer to section 2.

The processing performed by the Plastic Card Generation Server (SVPCGEN) is depicted in the following illustration.



Files

The files described below are required for BASE24-card processing.

- Cardholder Authorization File (CAF)
- Card Prefix File (CPF)
- Institution Definition File (IDF)
- Key Authorization File (KEYA)
- Logical Network Configuration File (LCONF)

Cardholder Authorization File (CAF)

The Cardholder Authorization File (CAF) contains one record for every cardholder whose card-issuing institution uses a positive identification method of authorization. Each record contains authorization parameters and usage accumulation information for the particular cardholder. The BASE24-card system accesses the CAF for the reissuing of new plastic cards for cardholder accounts.

Card Prefix File (CPF)

The Card Prefix File (CPF) contains one record for each card prefix in the system. Each prefix is defined according to its parent institution and includes corresponding Track 1 and Track 2 offsets, authorization criteria, PIN verification data, card verification data, and manual card verification data. The BASE24-card system accesses the CPF for cycle and production parameters for the reissuing of plastic cards.

Institution Definition File (IDF)

The Institution Definition File (IDF) contains one record for every card-issuing institution in the system. The IDF also contains holiday information for card usage accumulation statistics, the terminal cutover window times, PIN verification parameters, and sharing information. The BASE24-card system uses the IDF to reissue new plastic cards per institution.

Key Authorization File (KEYA)

The Key Authorization File (KEYA) contains one record for PIN verification, card verification digits (CVD), and manual CVD verification. The PIN, CVD, and manual CVD verification records for a card-issuing institution contain information needed by the BASE24-card system for PIN, PIN offset, CVD, and manual CVD generation during the issuing or reissuing of new plastic cards.

Logical Network Configuration File (LCONF)

The Logical Network Configuration File (LCONF) contains assign and param records for each required assign or param used by BASE24-card processes.

Soundex Algorithm

The Soundex algorithm is used by the list function of the Cardholder List Terminal (CLST) to access the Cardholder Authorization File (CAF) records using only a name. The CLST is designed for situations where an operator is not in possession of an account number but has at least an approximate spelling of a cardholder's name. The algorithm operates on a phonetic degree of similarity and provides a grouping of similar names and their corresponding card account numbers and member numbers from which the operator can choose.

This algorithm is also used to update the ALPHA-KEY field in the CAF. The ALPHA-KEY field is an alternate key to the CAF for use by the BASE24-card system when a cardholder name is added or amended on CAF screen 20, when CAF records are added or updated using the BASE24-from host maintenance product and when CAF records are added or updated using the Enscribe Refresh process.

Embossing, Encoding, and Indenting Overview

The BASE24-card product allows institutions the ability to configure information that is embossed, encoded, and indented onto plastic cards with a DataCard device. The embossing information is the data that is printed in raised characters on the card. The BASE24-card product supports up to four lines of emboss data. The encoding information is the data that is held in the magnetic stripe on the back of the card. The BASE24-card product supports Track 1, Track 2 and Track 3 encoding. The indenting information is the data that is indent printed in reverse-italic font on the back of the plastic card within the signature panel. The BASE24-card product supports one line of indent data. The information to be embossed, encoded, or indented is configured by card prefix on Cardholder Prefix File (CPF) screens 21 and 22.

The embossing, encoding, and indenting formats are described in detail in section 3.

Reporting

The BASE24-card product generates a comprehensive set of reports through which system activity can be tracked. The reports provide information needed by both managers and operators to supervise and maintain the BASE24-card system on a day-to-day basis, and to project the requirements for future activity.

The reports are broken down into two main categories. The categories, along with the reports they comprise, are described in the following paragraphs.

Production Control Reports

The Plastic Card Generation Server (SVPCGEN) produces production control reports. They consist of two reports that are always produced, and two optional reports. These reports are as follows:

Card Production Control Report — Enables the institution to retain a hardcopy of information about the plastic cards that were created.

Exception Report — Shows all exception conditions encountered during this processing run.

Card Carrier Report — An optional report, producing one card carrier per plastic card record processed, as indicated in the Card Prefix File and Cardholder Authorization File.

PIN Mailer Report — An optional report, producing one PIN mailer per plastic card record processed if software PIN mailer generation is required, as indicated in the Card Prefix File and Cardholder Authorization File.

Issue Reports

Issue reports provide information relating to the issue and reissue of plastic cards. They consist of the following four reports, each of which is produced by a different BASE24-card module:

Available Account Numbers Report/Labels — Generated by the Account Number Generation Server (SVAGEN)

CMS Reissue Projection Report — Generated by the issue projection function of the Card Management Function Server (SVC MF)

CMS Reissue Exception Report — Generated by the issue maintenance function of the Card Management Function Server (SVC MF)

CMS Issue Maintenance Report — Generated by the issue maintenance function of the Card Management Function Server (SVC MF)

A detailed description of each of the above-mentioned reports can be found in section 4.

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Section 2

Operations

This section describes all aspects of the BASE24-card operations. It is designed for use by individuals involved with card database maintenance and operating procedures for the card management system (CMS).

Files Maintenance

The BASE24-card product is a card issue and control system composed of two parts. One part is designed for daily database maintenance and system control functions using Pathway screens. The other part contains the Pathway screens used to launch card management functions and the programs that produce reports and output files for plastic card production.

This subsection contains a detailed description of each of the screens used for files maintenance. The BASE24-card product adds segments to two base files—the Cardholder Authorization File (CAF) and the Card Prefix File (CPF). The screens that access the BASE24-card segment of the CAF and CPF are described on the following pages.

The screens, programs, and procedures that perform card management production functions are described later in the Card Management Programs subsection.

BASE24 Virtual Menu

The BASE24 Virtual Menu provides access to all BASE24 files and functions. The BASE item accesses the Base Product Menu, from which you can select the CAF and CPF files maintenance screens. The CARD MANAGEMENT item accesses the Card Management Product Menu, from which you can select the following function screens used by the BASE24-card product:

- Card Management Functions (CMF)
- Card Operator Function (COF)
- Cardholder List Functions (CLST)

If an institution needs to control the issue of plastic cards, the Cardholder Authorization File (CAF) records, Card Prefix File (CPF) records, CMF, COF, and CLST are used to maintain the card information.

An example of the BASE24 Virtual Menu when the BASE24-card product is installed is shown below.

BASE24-ADMN	VIRTUAL MENU	LNET	YY/MM/DD	HH:MM	01 OF 01
V I R T U A L M E N U					
BASE			ATM		
POS			CARD MANAGEMENT		
CUSTOMER SERVICE			OPERATOR FUNCTIONS		
SWITCH FUNCTIONS					
***** BASE24 *****					
FILE DESTINATION:					
F1-ENTER DATA		F10-PRINT		F16-EXIT SF16-LOGOFF	

Cardholder Authorization File

Screen 20 of the Cardholder Authorization File (CAF) displays card management data for the BASE24-card product. This screen contains information about the cardholder and the cards issued to this cardholder.

When you add a CAF record through Pathway, the ALPHA-KEY field (an alternate key to the CAF, for specific use by the BASE24-card product) is added to the CAF record. This field is used to locate records using the Soundex algorithm. (The Soundex algorithm is described in section 1.) A subsequent execution of the LIST function of the Cardholder List Function (CLST) screen uses the ALPHA-KEY field from the CAF record. A subsequent execution of the ISSUE MAINTENANCE function from the Card Management Function (CMF) screen updates the GEN-KEY fields (another alternate key to the CAF, for specific use by the BASE24-card product) with the correct value to enable plastic card generation for the CAF record.

Screen 20 of the Cardholder Authorization File (CAF) is shown below, followed by descriptions of its fields.

BASE24-CMS	CARDHOLDER FILE	LLLL	YY/MM/DD	HH:MM	20 OF 20
PAN: _____		MEMBER: ____		FIID: ____	
CARD MANAGEMENT DATA					
NAME LINE1: _____		TITLE: _ ____			
NAME LINE2: _____		CITY: _____			
ADDR LINE1: _____		STATE: ____		ZIP CODE: _____	
ADDR LINE2: _____		COUNTRY: ____			
BRANCH NUMBER: ____		ISSUE #: ____			
DEPARTMENT NUMBER: ____		CARDS TO ISSUE: ____			
SERVICE CODE: ____		CARDS ISSUED: ____			
SECURITY CHARACTER: _		CARDS RETURNED: ____			
ISSUE DATE: _____		EFFECTIVE DATE: ____			
PIN MAILER REQUIRED: _ _____					
PLASTIC/CARD CARRIER OPTIONS: _ _____					
ISSUE STATUS: _ _____					
COMMENTS: _____					
***** BASE24 *****					
NEW PAGE: ____		FILE DESTINATION: ____		NEW LOGICAL NETWORK ID: ____	
F12 - HELP					

NAME LINE1 — The name of the primary cardholder. This field is required for card processing. This is the format of the embossed name on the card if the name is embossed on the card. The format of this field should be First Name**ℓ**Middle Initial**ℓ**Last Name, where **ℓ** represents a blank space.

TITLE — The title of the cardholder. Valid values are as follows:

- 0 = N/A
- 1 = Mr.
- 2 = Mrs.
- 3 = Miss
- 4 = Ms.
- 5 = Dr.

Default Value: 0

NAME LINE2 — The name of the secondary cardholder. This field is not required for card processing.

CITY — The city of residence for the cardholder. This field is required for card processing if PIN mailers or card carriers are to be created.

ADDR LINE1 — The first line of the cardholder's address. This field is required for card processing if PIN mailers or card carriers are to be created.

STATE — The state of residence for the cardholder. This field is required for card processing if PIN mailers or card carriers are to be created.

ZIP CODE — The ZIP code of residence for the cardholder. This field is required for card processing if PIN mailers or card carriers are to be created.

ADDR LINE2 — The second line of the cardholder's address.

COUNTRY — The ISO country code associated with the country of residence for the cardholder. Country codes are available in the ISO 3166:1993 standard, *Codes for the Representation of Names of Countries*.

Default Value: 000

BRANCH NUMBER — The branch identification of the institution issuing the plastic card. This field can be embossed on the card.

ISSUE # — The card issue number for this cardholder record. This field is incremented every time plastic card generation is performed for this cardholder and a plastic card, card carrier, or PIN mailer is produced.

Note: As long as one of these items is produced (card, carrier, or mailer) the value in this field is increased. However, it is increased just once each time plastic card generation is performed, even if multiple items are produced.

Default Value: 1

DEPARTMENT NUMBER — The department identification of the department issuing the plastic card. This field can be embossed on the card.

CARDS TO ISSUE — The number of plastic cards to issue for this cardholder on the next plastic card generation. This field should contain at least a value of 1 if a plastic card is to be produced.

Default Value: 1

SERVICE CODE — The ISO standard service code used to generate the card verification digits (CVD). This field can be embossed or encoded onto the plastic card.

Default Value: 0

CARDS ISSUED — The total number of plastic cards that have been issued to this cardholder. The value in this field is increased with the value from the CARDS TO ISSUE field during every plastic card generation run.

SECURITY CHARACTER — The security for the plastic card to be issued for the cardholder. This field is currently not used by the plastic card generation processing.

CARDS RETURNED — The total number of plastic cards returned to the institution for this cardholder.

Default Value: 0

ISSUE DATE — The date the plastic card generation function was performed for this cardholder.

EFFECTIVE DATE — The date the plastic card will become active. This field is set from the EFFECTIVE DATE field on Card Management Functions (CMF) screen 1 when a mass or force issue is performed and a plastic card is generated for the cardholder account.

PIN MAILER REQUIRED — A flag indicating whether the Plastic Card Generation Server (SVPCGEN) should create a PIN mailer for this cardholder. Valid values are as follows:

0 = No, do not produce a PIN mailer.

1 = Yes, produce a PIN mailer.

Default Value: 0

PLASTIC/CARD CARRIER OPTIONS — A flag indicating whether the Plastic Card Generation Server (SVPCGEN) process creates a plastic card or card carrier for this cardholder. Valid values are as follows:

0 = Plastic card only

1 = Both plastic card and card carrier

2 = Card carrier only

3 = No plastic card and no card carrier

Default Value: 0

ISSUE STATUS — The issue status for this cardholder record. This field is used to determine if a plastic card is produced for this cardholder. Valid values are as follows:

00 = Normal (no issue status)

01 = New issue

02 = Force issue

03 = Mass issue

04 = Expired

05 = Due to purge

07 = Due for reissue
09 = Suspend further issue
11 = New issue (in manufacture)
12 = Force issue (in manufacture)
13 = Mass issue (in manufacture)
17 = Reissue (in manufacture)

Default Value: 01

Note: If the ISSUE STATUS field is set to a value of 01, 02, 03, or 07, then the combination of PIN MAILER REQUIRED = 0 (No PIN mailer) and PLASTIC/CARD CARRIER OPTIONS = 3 (No plastic card and no card carrier) is not allowed.

COMMENTS — The institution can use this field for information about the cardholder account record. This field is updated by the value in the REASON field on CMF screen 1 when a force issue function is performed.

Card Prefix File

Card Prefix File (CPF) screens 20, 21, and 22 contain information used during issue maintenance and plastic card generation processing. Various cycles and production parameters are defined which allow an institution to control the number of reissued cards along with when and where these cards are produced. These parameters are used for all cards within this prefix and may be changed from time to time as an institution chooses.

Screen 20

Card Prefix File (CPF) screen 20 is shown below, followed by descriptions of its fields.

BASE24-CMS
PREFIX:

CARD PREFIX

LLLL
PAN LENGTH: 00

YY/MM/DD
FIID:

20 OF 22

CARD MANAGEMENT DATA

PROCESSING INFORMATION

VENDOR TYPE: A

STOCK TYPE: A

LENGTH OF ISSUE: 12 (MONTHS)

REISSUE CYCLE: 30 (DAYS)

CHECK DIGIT ALGO: 1 (LUHN)

PRODUCTION CYCLE: 15 (DAYS)

PIN LENGTH: 4

EXPIRATION CYCLE: 0 (DAYS)

PIN GENERATION TYPE: N (NATURAL)

PIN MAILER CONTROL: 0 (NO PIN MAILER)

CARD CARRIER CONTROL: 0 (NO CARD CARRIER)

LAST ACCOUNT NUMBER: 00000000000000000000000000000000

ISSUE CONTROL DATA

CARD STATUS: 00 (NORMAL STATUS)

TAPE INFORMATION

TAPE LABEL TYPE: NON (NO LABELS)

***** BASE24 *****

NEW PAGE:

FILE DESTINATION:
F12-HELP

NEW LOGICAL NETWORK ID:

PROCESSING INFORMATION

The following fields are used to control the reissuing of plastic cards and card control information. These fields are used during the Issue Maintenance and Plastic Card Generation processing.

VENDOR TYPE — This field indicates the manufacturer of the plastic cards to be used for card generation. This field is used for plastic card generation processing to create the disk output file for the plastic cards to be created. A separate output file is used for each vendor type regardless if the output is sent to tape or disk. This ensures that the correct plastic cards are loaded into the card embossing device before the file is processed. Valid values are A–Z.

Default Value: A

STOCK TYPE — The type of plastic from the manufacturer to be used for card generation.

This field is used during plastic card generation processing to create the disk output file for the plastic cards to be created. A separate disk output file is used for each stock type processed. This ensures that the correct plastic cards are loaded in to the card embossing device before the file is processed. Valid values are A–Z.

Default Value: A

LENGTH OF ISSUE — The number of months the card will be active. This field is used during issue maintenance processing to determine the new expiration date when a card is due to be reissued. Valid values are 6–192 months, in multiples of 6.

Default Value: 12 (months)

REISSUE CYCLE — The number of days before expiration a cardholder account can be reviewed to reissue a new plastic card. This field is used during issue maintenance processing to determine if a card can be reissued. Valid values are 30–120 days, in multiples of 15.

Default Value: 30 (days)

CHECK DIGIT ALGO — The check digit value used in generating the Available Account Numbers Report. The BASE24-card product supports only the value of 1 (Luhn).

Default Value: 1

PRODUCTION CYCLE — The number of days before expiration a cardholder account can be reviewed for production. This field is used during issue maintenance processing to determine if a card can be produced. Valid values are 15–45 days, in multiples of 15.

Default Value: 15 (days)

PIN LENGTH — The number of digits in the PIN for a card with this prefix. Valid values are 4–12.

Variable length PINs are supported only with the Thales e-Security (Racal) HSM (version 7000) using IBM DES PIN generation. Software PIN generation and ABA Visa PIN generation using the Thales e-Security (Racal) HSM must use four-digit PINs.

Default Value: 4

EXPIRATION CYCLE — The number of days before this record will be automatically purged from the system if the card has expired. This field is used during issue maintenance processing to automatically purge expired records from the CAF. Valid values are as follows:

0 = Do not purge expired records

1–9999 = Days before expired records will be purged

Default Value: 0

PIN GENERATION TYPE — A flag indicating whether random or natural PIN generation is used for this prefix. Valid values are as follows:

N = Natural PIN generation (PIN offset of zeros)

R = Random PIN generation (with a PIN offset)

The Thales e-Security (Racal) HSM using IBM DES PIN generation supports both natural PIN generation and random PIN generation. The Thales e-Security (Racal) HSM using ABA Visa PIN generation only supports random PIN generation. Software PIN generation only supports random PIN generation.

Default Value: N

PIN MAILER CONTROL — A flag indicating whether a PIN mailer is created for the Cardholder Authorization File (CAF) records with this prefix. Valid values are as follows:

0 = No PIN mailer

1 = PIN mailer on Tandem processor

2 = PIN mailer on card machine

The current release of the BASE24-card product does not support printing PIN mailers on the card machine. Therefore, if a value of 2 is entered for this field, it is processed as a value of 0 (No PIN mailer).

Default Value: 0

CARD CARRIER CONTROL — A flag indicating whether a card carrier is created for the Cardholder Authorization File (CAF) records with this prefix. Valid values are as follows:

0 = No card carrier

1 = Card carrier on Tandem processor

2 = Card carrier on card machine

The current release of the BASE24-card product does not support printing card carriers on the card machine. Therefore, if a value of 2 is entered for this field, it is processed as a value of 0 (No card carrier).

Default Value: 0

LAST ACCOUNT NUMBER — The last available account number generated by the Available Account Numbers Report. This field is used and updated by the Account Number Generation process to determine the next possible account number for the Available Account Numbers Report.

Default Value: 00000000000000000000000000000000

ISSUE CONTROL DATA

The following field indicates which cardholder accounts can be issued for this prefix.

CARD STATUS — The valid status of cardholder accounts (from the CARD STATUS field on CAF Screen 1) for which accounts can be issued for this prefix. This field is used by the issue maintenance and plastic card generation processes to determine if a plastic card is issued for a cardholder account. Valid values are as follows:

- 00 = Issue cards with normal status
- 01 = Issue cards with normal or hot status
- 02 = Issue cards with normal or inactive status
- 03 = Issue cards with normal or closed status
- 04 = Issue cards with normal, hot, or inactive status
- 05 = Issue cards with normal, hot, or closed status
- 06 = Issue cards with normal, inactive, or closed status
- 07 = Issue cards with normal, hot, inactive, or closed status

The BASE24-card product does not issue or reissue a card with a lost or stolen (hot) status for security reasons. Therefore, the values of 01, 04, 05, and 07 do not allow cards to be issued or reissued if the account has a hot status. Refer to the Card Production Examples subsection for an example of issuing a lost or stolen card.

Default Value: 00

TAPE INFORMATION

The following field is used to indicate which type of tape label to use when processing to tape.

TAPE LABEL TYPE — The type of label to be written to the tape for plastic card generation when the output device is a tape. Valid values are as follows:

- BUR = Burroughs label
- IBM = IBM label
- NON = Nonlabeled
- DOS = IBM DOS Labels

Default Value: NON

Screen 21

Screen 21 of the Card Prefix File (CPF) defines embossing and encoding control information. Screen 21 is shown below, followed by descriptions of its fields.

BASE24-CMS
PREFIX:

CARD PREFIX

LLLL
PAN LENGTH: 00

YY/MM/DD
FIID:

21 OF 22

CARD MANAGEMENT DATA

EMBOSSING CONTROL
FORMAT TYPE: 00 (DEFAULT, WORK-MBR)
LINE 1 FORMAT:
LINE 2 FORMAT:
LINE 3 FORMAT:
LINE 4 FORMAT:

ENCODING CONTROL:
FORMAT TYPE: 00 (DEFAULT FORMATTING)
TRACK2 DISCRETIONARY DATA FORMAT:

TRACK 1 FORMAT:

***** BASE24 *****

NEW PAGE:FILE DESTINATION:NEW LOGICAL NETWORK ID:
F12-HELP

EMBOSSING CONTROL

The following fields are used to indicate what information is embossed on the plastic card. The Plastic Card Generation Server (PCGEN) uses these fields to configure the embossing information. For more information on embossing formats, refer to section 3.

FORMAT TYPE — The type of format used to control the information embossed on the plastic card. Valid values are as follows:

- 00 = Use the default format + Work-Mbr
- 01 = Use template formatting
- 02 = Use the default format + EXP + Work-Mbr + CVV value
- 03 = Use the default format only

The default format is described in detail in section 3.

Default Value: 00

LINE 1 FORMAT — This field will be filled with the template values for embossing information on the first emboss line of the plastic card if the FORMAT TYPE field is set to a value of 01.

LINE 2 FORMAT — This field will be filled with the template values for embossing information on the second emboss line of the plastic card if the FORMAT TYPE field is set to a value of 01.

LINE 3 FORMAT — This field will be filled with the template values for embossing information on the third emboss line of the plastic card if the FORMAT TYPE field is set to a value of 01.

LINE 4 FORMAT — This field will be filled with the template values for embossing information on the fourth emboss line of the plastic card if the FORMAT TYPE field is set to a value of 01.

ENCODING CONTROL

The following fields are used to indicate what information is encoded on the magnetic stripe of the plastic card. The Plastic Card Generation Server (SVPCGEN) uses these fields to configure the encoded information. For more information on encoding formats, refer to section 3.

FORMAT TYPE — The type of format used to control the information encoded on the magnetic stripe of the plastic card. Valid values are as follows:

- 00 = Use the default format
- 01 = Use template formatting

The default format is described in detail in section 3.

Default Value: 00

TRACK2 DISCRETIONARY DATA FORMAT — This field is filled with the template values for encoding information on TRACK2 of the plastic card if the FORMAT TYPE field is set to a value of 01.

TRACK1 FORMAT — This field is filled with the template values for encoding information on TRACK1 of the plastic card if the FORMAT TYPE field is set to a value of 01.

Screen 22

Screen 22 of the Card Prefix File (CPF) defines encoding controls for Track 3 information as well as indent printing controls. Screen 22 is shown below, followed by descriptions of its fields.

BASE24-CMS PREFIX:	CARD PREFIX	LLLL PAN LENGTH: 00	YY/MM/DD FIID:	HH:MM 22 OF 22
CARD MANAGEMENT DATA				
ENCODING CONTROL (TRACK3 CONSTANT FIELDS)				
FORMAT CODE: 01 (TRACK3 ENCODED)				
COUNTRY CODE: 840	CURRENCY CODE: 840	CURRENCY EXPONENT: 0		
CYCLE LENGTH: 00 (DAYS)	RETRY COUNT: 0	ALGO ID: 00		
INTRCHG CNTL: 0	RELAY MARKER: 0	ATALLA OFFSET: 000000		
ACCT LENGTH: 00	DFLT PRI TYP: 00	DFLT DDA SAN TYP: 00		
DFLT SAV SAN TYP: 00	DFLT NO SAN TYP: 00			
INDENT PRINTING CONTROL:				
FORMAT TYPE: 00 (NO INDENT PRINTING)				
INDENT FORMAT:				
***** BASE24 *****				
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:		
	F12-HELP			

ENCODING CONTROL (TRACK3 CONSTANT FIELDS)

The following fields are used to indicate the information encoded on Track 3 of the magnetic stripe of the plastic card. For more information on the format of Track 3, please refer to the ISO 4909 standard, *Magnetic Stripe Data Content for Track 3*.

FORMAT CODE — A code indicating whether or not Track 3 data is encoded on the magnetic stripe. Valid values are as follows:

00 = Track 3 not encoded
01 = Track 3 encoded

Default Value: 01

COUNTRY — The ISO country code associated with the country of residence for the institution. Country codes are available in the ISO 3166:1993 standard, *Codes for the Representation of Names of Countries*.

Default Value: 840

CURRENCY CODE — A code indicating the currency used when calculating for the update. Valid values are listed in the ISO 4217:1995 standard, *Codes for the Representation of Currencies and Funds*.

Default Value: 840

CURRENCY EXPONENT — A digit indicating the number of times the amount authorized and amount remaining must be multiplied by 10 to express the value in the major currency unit.

Default Value: 0

CYCLE LENGTH — A code representing the period of time in days during which the accumulated sum of all debit transactions shall not exceed the amount authorized. Valid values are 00–99 days.

Default Value: 00

RETRY COUNT — The number of incorrect attempts available to enter the personal identification number (PIN) associated with this card. Valid values are 0–9.

Default Value: 0

ALGO ID — A code identifying the algorithm to be used. Valid values are as follows:

00–09 = Private algorithm
10–19 = DEA-1

Default Value: 00

INTRCHG CNTL — A flag indicating whether interchange is permitted on this card. Valid values are as follows:

0 = No restriction.
1 = Not available for international interchange.
2–8 = Interchange restricted.
9 = Interchange restricted; recommended for test cards.

Default Value: 0

RELAY MARKER — A flag indicating whether the interchange message is to include the contents of the additional data field. Valid values are as follows:

- 0 = Include all discretionary data.
- 1 = Do not include additional data.
- 2 = Do not include discretionary data.

Default Value: 0

ATALLA OFFSET — A six-digit field, which provides a means of verifying the integrity of the data elements on Track 3 through application of a cryptographic formula. Valid values are 000000–999999.

Default Value: 000000

ACCT LENGTH — The length of the subsidiary account number. Valid values are 00–28.

Default Value: 00

DFLT PRI TYP — A code identifying the type of account and service restriction for the primary account. Valid values are as follows:

Type of account—first digit

- 0 = PAN not encoded on Track 3
- 1 = Savings account
- 2 = Current or checking account
- 3 = Credit card account
- 4 = Universal account number
- 5 = Interest-bearing current or checking account

Service restrictions—second digit

- 0 = No restrictions
- 1 = No cash dispense
- 2 = No point of service (POS) transaction
- 3 = No cash dispense and no POS transaction
- 4 = Positive authorization required

Default Value: 00

DFLT DDA SAN TYP — A code identifying the type of account and service restriction for the checking account. Valid values are as follows:

Type of account—first digit

- 0 = SAN-1 not encode on Track 3
- 1 = Savings account
- 2 = Current or checking account
- 3 = Credit card account
- 4 = Universal account number
- 5 = Interest-bearing current or checking account

Service restrictions—second digit

- 0 = No restrictions
- 1 = No cash dispense
- 2 = No point of service (POS) transaction
- 3 = No cash dispense and no POS transaction
- 4 = Positive authorization required

Default Value: 00

DFLT SAV SAN TYP — A code identifying the type of account and service restriction for the savings account. Valid values are as follows:

Type of account—first digit

- 0 = SAN-2 not encode on Track 3
- 1 = Savings account
- 2 = Current or checking account
- 3 = Credit card account
- 4 = Universal account number
- 5 = Interest-bearing current or checking account

Service restrictions—second digit

- 0 = No restrictions
- 1 = No cash dispense
- 2 = No point of service (POS) transaction
- 3 = No cash dispense and no POS transaction
- 4 = Positive authorization required

Default Value: 00

DFLT NO SAN TYP — A code identifying the type of account and service restriction when there are no accounts associated with this PAN. This field should be set to a value of 00 (SAN not encoded on Track 3).

Default Value: 00

INDENT PRINTING CONTROL

The following fields are used to indicate what information is to be indented on the plastic card. The Plastic Card Generation Server (SVPCGEN) process uses these fields to configure the indenting information. For more information on the Indenting formats, refer to section 3.

FORMAT TYPE — The type of format used to control the information indented on the plastic card. Valid values are as follows:

- 00 = No indent printing
- 01 = Use template formatting
- 02 = Use the default format (CVV2 only)

Default Value: 00

The default format is described in detail in section 3.

INDENT FORMAT — This field is filled with the template values for indenting information on the back of the plastic card within the signature panel, if the FORMAT TYPE field is set to a value of 01.

Card Management Programs

The BASE24-card product provides institutions with the ability to control the issuing and maintenance of plastic cards on the Tandem computer system. The files maintenance segment was described earlier in the Files Maintenance subsection. This subsection describes the screens, programs, and procedures for producing reports and output files for plastic card production.

BASE24-card Product Menu

The following screen print is of the BASE24-card Product Menu. From this menu, you can access the following screens:

- The Card Management Functions (CMF) screen generates plastic cards, performs issue maintenance, and generates the CMS Reissue Projection Report.
- The Cardholder List Function (CLST) screen locates a cardholder's account number by entering the cardholder's name.
- The Card Operator Function (COF) screen generates the Available Account Numbers Report.

```
BASE24-ADMN  MENU                LLLL      YY/MM/DD  HH:MM  01 OF 01
                        C M S      P R O D U C T  M E N U
                        CLST      CMF      COF

***** BASE24 *****
                        FILE DESTINATION:
F1-ENTER DATA  F9-NEXT PAGE  F11-PREVIOUS PAGE  F10-PRINT  F12-HELP
```

Card Management Functions

The Card Management Functions (CMF) screen allows the operator to perform the following functions:

- Issue projection
- Issue maintenance
- Force issue (initiates the plastic card generation process)
- Mass issue (initiates the plastic card generation process)

Card Management Functions (CMF) screen 1 is shown below, followed by descriptions of its fields.

```

BASE24-CMF   CARD MANAGEMENT           LNET           YY/MM/DD  HH:MM  01 OF 01
FIID: _____

FORCE ISSUE DATA:
  CARD NUMBER _____ PREFIX LENGTH ____
  REASON _____
  EFFECTIVE DATE _____ (YYMMDD OR LEAVE BLANK TO USE TODAYS DATE)

MASS ISSUE DATA: PREFIX _____ PAN LENGTH ____
  EFFECTIVE DATE _____ (YYMMDD OR LEAVE BLANK TO USE TODAYS DATE)

OUTPUT DEVICE FOR FORCE/MASS ISSUE _ (T=TAPE D=DISK E=EMBOSS MACHINE)

PRESS F2 TO START FORCE ISSUE / PRESS F4 TO START ISSUE PROJECTION PROCESS
PRESS F3 TO START MASS ISSUE / PRESS F5 TO START ISSUE MAINTENANCE PROCESS

***** BASE24 *****
NEW PAGE:  _ FILE DESTINATION: _____ NEW LOGICAL NETWORK ID: _____
F1-ENTER  _ F10-PRINT  F12-HELP  F16-EXIT

```

FIID — The FIID of the institution for which the plastic card management function is being performed. This field is required and wildcard values are not supported.

FORCE ISSUE DATA

The following fields are used when performing a force issue function to individually create a plastic card.

CARD NUMBER — The number of the plastic card being force issued.

PREFIX LENGTH — The length of the prefix for this cardholder account record being force issued. Valid values are 0–11.

REASON — The reason the plastic card must be force issued. This field is required on a force issue. This field will update the COMMENTS field on CAF screen 20 for the corresponding CAF record when a force issue is performed. Valid values are any alphanumeric characters.

EFFECTIVE DATE — The date this force-issued card will become active. This field updates the EFFECTIVE DATE field on CAF screen 20 for the corresponding CAF record when a force issue is performed. If no date is entered in this field, the current date is used as the effective date for the card created. The format for this field is YYMMDD.

MASS ISSUE DATA

The following fields are used when performing a mass issue function to create all of the plastic cards on the BASE24 system that require generation.

PREFIX — Indicates whether the mass issue function is performed for a specific prefix or all prefixes for an FIID. If you enter a prefix value in this field, all of the plastic cards that need to be created are generated for the specified prefix. If you leave this field blank, all of the plastic cards that need to be created are generated for the FIID.

PAN LENGTH — The length of the primary account number (PAN) as it appears in the CPF record for the above-mentioned prefix. This field must be entered if a prefix value is entered in the PREFIX field.

EFFECTIVE DATE — The date the mass-issued cards become active. The value from this field is used to update the EFFECTIVE DATE field on CAF screen 20 for the corresponding CAF records processed. If no date is entered in this field, the current date is used as the effective date for the cards created. The format for this field is YYMMDD.

OUTPUT DEVICE FOR FORCE / MASS ISSUE — A flag indicating the output device to be used for a force or mass issue. Valid values are as follows:

D = Tandem processor disk
T = Tape
E = Embossing machine

This field is required when performing a force issue or mass issue function.

Issue Projection Function

The issue projection function generates the CMS Reissue Projection Report, which contains a month-by-month total of cards that are due for reissue. The report is used to forecast the supplies needed for card production and allows the institution to establish a viable timetable for reissuing cards. See the CMS Reissue Projection Report in the Card Reporting section of this manual for more information concerning the interpretation of this report.

A valid FIID must appear in the FIID field at the top of the CMF screen in order to perform the issue projection function.

The following Logical Network Configuration File (LCONF) assigns are required to run the issue projection process:

CPF — The Tandem file location of the Card Prefix File (CPF).

IDF — The Tandem file location of the Institution Definition File (IDF).

ISSUE-PROJ-RPT — The location of the CMS Reissue Projection Report. This location can be a Tandem spooler location or the name of a Tandem disk file.

You can generate this report at any time from the CMF screen, by performing the following steps:

1. Enter the FIID of the institution to be reported on in the FIID field.
2. Press the **F4** key to start the issue projection report process.

Issue Maintenance Function

The issue maintenance function performs a variety of tasks, all of which involve examining the individual card accounts to determine whether the card account is reissued during the next plastic card generation run.

This function is required before running the plastic card generation process (i.e., before performing a force or mass issue function). If this function is not executed before performing a force or mass issue function, none of the cards requiring generation is produced.

The following LCONF assigns are required to run the issue maintenance process:

CPF — The Tandem file location of the Card Prefix File (CPF).

EXCEPTION-RPT — The location of the CMS Reissue Exception Report. This location can be a Tandem spooler location or a Tandem disk file.

IDF — The Tandem file location of the Institution Definition File (IDF).

ISSUE-MAINT-RPT — The location of the CMS Issue Maintenance Report. This location can be a Tandem spooler location or the name of a Tandem disk file.

You can start the issue maintenance function from the CMF screen by performing in the following steps:

1. Enter the FIID of the institution to perform the issue maintenance on in the FIID field.
2. Press the **F5** key to begin the issue maintenance function.

Force Issue Function

The BASE24-card system allows plastic cards to be individually produced in emergency situations. When a card needs to be issued immediately, it must be flagged and included in the Plastic Card Generation Server (SVPCGEN) output file. This is accomplished through the force issue function.

The force issue function starts the Plastic Card Generation Server (SVPCGEN) process to create the card production output file. An issue maintenance function is first performed for the FIID of the card to be issued, which sets the GEN-KEY field of the card account record in the CAF to indicate that the card account must be issued at the next SVPCGEN run. When the force issue function is then performed for the desired card, the only record in the SVPCGEN output file will be for this card number.

For a listing of the LCONF assigns and params required to perform the force issue function, refer to the topic “Plastic Card Generation Server Process” later in this section.

You can force issue an individual plastic card from the CMF screen by performing the following steps:

1. Enter the FIID, card account number, prefix length, reason and effective date (if different from the current date) in the appropriate fields.
2. Enter the appropriate output device value in the OUTPUT DEVICE FOR FORCE/ MASS ISSUE field. If the output device is to tape, proceed with step 3. If the output device is to disk, skip to step 6.

3. From a TACL session, enter the following command:

PATHCOM \$XXXX

Where \$XXXX is the PATHMON process name defined in the PATHCOOL or PATHCOLD obey files.

4. From the Pathcom prompt, enter the following command:

=ALTER SERVER-PCGEN, HOMETERM \$XXXXXXXXX

Where \$XXXXXXXXX is the name of the terminal to be used as the interactive device. This device must be “paused out” in order for the operator to receive the tape mount messages from SVPCGEN.

5. Exit from Pathcom by entering the following command:

=EXIT

6. Press the **F2** key to begin the force issue process. If the output device is to tape, proceed with step 7. If the output device is to disk, there are no more steps required.

7. The prompt to mount the magnetic tape will appear on the terminal that has been defined as the interactive device in step 4. Respond to the prompt by pressing ENTER.

8. When the SVPCGEN process displays a completion message, start another Pathcom session by entering the same command entered in step 3.

PATHCOM \$XXXX

9. From the Pathcom prompt, enter the following command:

=ALTER SERVER-PCGEN, HOMETERM \$SUDO

All messages from the SVPCGEN process will now be logged through the Event Management Service (EMS).

10. Exit from Pathcom by entering the following command:

=EXIT

Mass Issue Function

The BASE24-card system enables institutions to produce cards on a periodic basis for each FIID on the system. The mass issue function allows the operator to generate the plastic cards for reissues and new issues for the institution.

You can run the mass issue function for a single prefix, or for the whole FIID. The issue maintenance function must be performed for the FIID prior to performing the mass issue function. The mass issue function starts the SVPCGEN process to create the output file for card generation. When the mass issue function is performed, the output file will contain all cards for the FIID or prefix that need to be generated. For a listing of the LCONF assigns and params required to perform the mass issue function, refer to the topic “Plastic Card Generation Server Process” later in this section.

You can mass issue plastic cards from the CMF screen by performing the following steps:

1. Enter the FIID, prefix (if only issuing for a specific prefix), PAN length and effective date (if other than the current date) in the appropriate fields.
2. Enter the appropriate output device value in the OUTPUT DEVICE FOR FORCE/ MASS ISSUE field. If the output device is to tape, proceed with step 3. If the output device is to disk, skip to step 6.

3. From a TACL session, enter the following command:

```
PATHCOM $XXXX
```

Where \$XXXX is the PATHMON process name defined in the PATHCOOL or PATHCOLD obey files.

4. From the Pathcom prompt, enter the following command:

```
=ALTER SERVER-PCGEN, HOMETERM $XXXXXXXXXX
```

Where \$XXXXXXXXXX is the name of the terminal to be used as the interactive device. This device must be “paused out” in order for the operator to receive the tape mount messages from SVPCGEN.

5. Exit from Pathcom by entering the following command:

```
=EXIT
```

6. Press the **F3** key to begin the mass issue process. If the output device is to tape, proceed with step 7. If the output device is to disk, there are no more steps required.
7. The prompt to mount the magnetic tape will appear on the terminal that has been defined as the interactive device in step 4. Respond to the prompt by pressing ENTER.
8. When the SVPCGEN process displays a completion message, start another Pathcom session by entering the same command entered in step 3.

```
PATHCOM $XXXX
```

9. From the Pathcom prompt, enter the following command:

```
=ALTER SERVER-PCGEN,HOMETERM $SUDO
```

All messages from the SVPCGEN process will now be logged through the Event Management Service (EMS).

10. Exit from Pathcom by entering the following command:

```
=EXIT
```

Plastic Card Generation Server Process

The Plastic Card Generation Server (SVPCGEN) process creates an output file (tape or disk), unless outputting directly to the DataCard machine, that contains all the information necessary for the plastic card vendor to emboss, encode, and indent plastic cards. A separate output file is also created for a PIN Mailer Report and Card Carrier Report.

SVPCGEN is initiated using the force issue or mass issue function from the CMF screen and should be run periodically to generate the cards on the system that require creation.

The following LCONF assigns are required to run the SVPCGEN process:

CARD-CARRIER-RPT — The location of the Card Carrier Report. This location can be a Tandem spooler location or the name of a Tandem disk file.

CARD-PRODUCTION-RPT — The location of the Card Production Control Report. This location can be a Tandem spooler location or the name of a Tandem disk file.

PCGEN-EXCEPTION-RPT — The location of the PCGEN Exception Report. This location can be a Tandem spooler location or the name of a Tandem disk file.

PIN-MAILER-RPT — The location of the software PIN Mailer Report. This location can be a Tandem spooler location or the name of a Tandem disk file.

CPF — The Tandem disk file location of the Card Prefix File (CPF).

EMBOSS-MACHINE — The Tandem port name to which the DataCard machine is attached. This assign is only required if the output device for plastic card generation is the DataCard embossing machine.

TAPE-DRIVE — The Tandem tape drive name for which BASE24 is to write the card generation data. This assign is only required if the output device for plastic card generation is tape.

IDF — The Tandem disk file name of the Institution Definition File (IDF).

KEYA — The Tandem disk file name of the Key Authorization File (KEYA).

SECURE-DEV1 — The Tandem port name for the primary Hardware Security Module (HSM). If the HSM is a Thales e-Security (Racal), the User Field of this LCONF assign record must contain RG6000 or RG7000 to indicate whether a Version 6000 or Version 7000 device is being used. (The Version 7000 device accepts a PIN-Length parameter to generate a variable-length PIN. The Version 6000 device only creates four-digit PINs.) If the User Field is blank, SVPCGEN will default to a Version 7000 device.

Note: The SVPCGEN process attempts to read this LCONF assign with a process name of SVPCGEN. If no match is found, the process attempts to find this assign with a default process name of “*****”.

SECURE-DEV2 — The Tandem port name for the secondary Hardware Security Module (HSM). If the HSM is a Thales e-Security (Racal), the User Field of this LCONF assign record must contain RG6000 or RG7000 to indicate whether a Version 6000 or Version 7000 device is being used. (The Version 7000 device accepts a PIN-Length parameter to generate a variable-length PIN. The Version 6000 device only creates four-digit PINs.) If the User Field is blank, SVPCGEN will default to a Version 7000 device.

Note: The SVPCGEN process attempts to read this LCONF assign with a process name of SVPCGEN. If no match is found, the process attempts to find this assign with a default process name of “*****”.

The following LCONF param is required to run the SVPCGEN process:

SECURE-DEV-TYP — The type of Hardware Security Module (HSM).

Note: The SVPCGEN process attempts to read this LCONF param with a process name of SVPCGEN. If no match is found, the process attempts to find this param with a default process name of “*****”.

The following LCONF params are optional to run the SVPCGEN process:

HRDWR-CVV-GEN — A flag indicating whether the Card Verification Value (CVV) and Card Verification Value 2 (CVV2) are generated in hardware. If this param is set to a value of Y, the CVV and CVV2 are generated in hardware. If this param is missing or set to any other value, the CVV and CVV2 are generated in software.

HRDWR-PIN-GEN — A flag indicating if the PIN and PIN mailer are generated in hardware. If this param is set to a value of Y, the PIN and PIN mailer are generated in hardware. If this param is missing or set to any other value, the PIN and PIN mailer are generated in software.

Cardholder List Functions

The Cardholder List (CLST) screen is used to locate a cardholder's account number by entering an approximate spelling of the cardholder's name.

Cardholder List (CLST) screen 1 is shown below, followed by descriptions of its fields.

BASE24-CMS	CARDHOLDER LIST	LN	NET	YY/MM/DD	HH:MM	01 OF 01
Enter Name				FIID		Member #
Name and Address				Card #		
-						
-						
-						
-						
-						
-						
-						
-						
-						
-						
-						
***** BASE24 *****						
FILE DESTINATION: _____ NEW LOGICAL NETWORK ID: _____						
F1-LIST	F2-SELECT	F9-NEXT PAGE	F10-PRINT	F12-HELP	F16-EXIT	

ENTER NAME — The name of the cardholder for inquiry in the format entered in the NAME LINE 1 field on CAF screen 20. (If the exact spelling of the cardholder's name is known, enter it in the following format: First Name**b**Middle Initial**b**Last Name, where **b** represents a blank space. Otherwise, enter an approximate spelling of the cardholder's last name.)

FIID — The FIID of the cardholder for inquiry. This field is required when performing a list function.

NAME AND ADDRESS — These fields display all cardholder names and addresses phonetically matching the cardholder name entered in the ENTER NAME field for the specified FIID.

CARD # — These fields display all card numbers for the listed cardholders.

MBR # — These fields display all member numbers for the listed cardholders.

The CLST allows the operator to get a listing of cardholders by entering an approximate spelling of the name as it appears in the NAME LINE 1 field of CAF screen 20. The Soundex algorithm creates a phonetic spelling of the name entered in the ENTER NAME field on CLST screen 1 that is compared to the ALPHA-KEY field values of the corresponding CAF for the FIID entered. Once the list function displays all of the possible cardholders, the operator can select the correct one and CAF screen 1 is displayed with the record selected.

The following LCONF assign is required to run the cardholder list function:

IDF — The Tandem file location of the Institution Definition File (IDF).

You can list cardholders from the CLST screen by performing the following steps:

1. Enter an approximate spelling of the cardholder's name in the ENTER NAME field in the format of the NAME LINE 1 field on CAF screen 20 (e.g., SMITH).
2. Enter the FIID for the cardholder inquiry in the FIID field (e.g., BNK2).
3. Press the **F1** key to perform the list function.

All of the cardholders for the specified FIID whose last name is phonetically similar to name in the ENTER NAME field are listed on the CLST screen (e.g., SMYTH, SMITHE, etc.).

4. To select one of the records shown, place the cursor next to the cardholder name and press the **F2** key. The CAF record for the selected cardholder is retrieved and displayed.

Card Operator Function

The Card Operator Function (COF) screen allows the operator to generate the Available Account Numbers Report for a specified prefix.

Card Operator Function (COF) screen 1 is shown below, followed by a description of its fields.

BASE24-CMS	OPERATOR FUNCTIONS	LNEXT	YY/MM/DD	HH:MM	01 OF 01
FIID: _____					
ACCOUNT NUMBER GENERATION DATA					
PREFIX: _____		PREFIX LENGTH: ____	PAN LENGTH: ____		
OUTPUT TYPE: ____ (R=REPORT, L=LABEL)					
NUMBER OF ACCOUNTS: _____					
***** BASE24 *****					
NEW PAGE: ____		FILE DESTINATION: _____	NEW LOGICAL NETWORK ID: _____		
F1-ENTER		F2-ACCT NO GEN	F12-HELP	F16-EXIT	

FIID — The FIID for which the Account Number Generation Report is to be run.

ACCOUNT NUMBER GENERATION DATA

The following fields are used to control the generation of the Available Account Numbers Report.

PREFIX — The prefix for which the Available Account Numbers Report is to be run.

PREFIX LENGTH — The length of the requested prefix.

PAN LENGTH — The length of the card numbers for the requested prefix.

OUTPUT TYPE — A flag indicating the type of output needed for the Account Number Generation function. Valid values are as follows:

R = Report

L = Labels

NUMBER OF ACCOUNTS — The number of possible account numbers to be generated for the requested prefix. Valid values are 0–9999.

The COF screen is used to enter the prefix information of the possible account numbers to be generated, the type of production desired (labels or listing) and the number of the available

account numbers to be listed.

The following LCONF assigns are required to run the Account Number Generation (SVAGEN) process:

ACCOUNT-NUM-RPT — The location of the Available Account Numbers Report. This location can be a Tandem spooler location or the name of a Tandem disk file.

CPF — The Tandem disk file location for the Card Prefix File (CPF).

After the parameters have been entered on COF screen 1, press the **F2** key to begin the Account Number Generation (SVAGEN) process. The account numbers are generated sequentially and written to the report location specified in the ACCOUNT-NUM-RPT assign in the label or listing format. When all of the numbers have been generated, the LAST ACCOUNT NUMBER on CPF screen 20 is updated to reflect the last account number generated.

For more information on the Available Account Number Report, refer to section 4.

Card Production Examples

The BASE24-card system enables an institution to control the production of plastic cards for multiple customer bases.

There are many ways to accomplish the production of plastic cards, PIN mailers and card carriers. To help in understanding the available options, some typical examples of card production needs are detailed in this subsection.

The following general configuration issues apply to all of the examples provided in this subsection:

- Depending on the PIN generation method (hardware or software), the type of Hardware Security Module (HSM), and the PIN encryption method (IBM DES or ABA Visa), different PIN types (natural, random, and four-digit, variable length) can be implemented.
- A random PIN is generated with a PIN offset (IBM DES) or a PVV (ABA Visa). The PIN offset or PVV is created with the same number of digits as the PIN.
- A natural PIN is generated with a PIN offset of zeros.
- Random PINs can be created with software PIN generation or with the Thales e-Security (Racal) HSM (version 6000 or 7000) using IBM DES or ABA Visa PIN generation.
- Natural PINs can be created only with the Thales e-Security (Racal) HSM (version 6000 or 7000) using IBM DES PIN generation.
- Four-digit PINs can be created with software PIN generation or with the Thales e-Security (Racal) HSM (version 6000 or 7000) using IBM DES or ABA Visa PIN generation.
- Variable-length PINs (random or natural) can be created only with the Thales e-Security (Racal) HSM (version 7000) using IBM DES PIN generation.

Example One—New Account

A card administrator needs to add a new account to the card database maintained within BASE24, and have a plastic card produced.

The card administrator logs on to the BASE24 CRT access system and proceeds to the Cardholder Authorization File (CAF). The card administrator enters the cardholder's account

information along with the name and address where the plastic card, PIN mailer and card carrier are to be sent, and an issue status of 01 (New issue).

When the card administrator adds the record to the CAF, the ALPHA-KEY alternate key is created. This field allows the Cardholder List Function (CLST) to be performed on this account.

A system operator performs the issue maintenance function at the end of the day or end of the week (this is determined by the institution) to prepare all the cardholder accounts for plastic card generation. When the issue maintenance function is executed, another alternate key called the GEN-KEY is created for all new issue cards. A subsequent run of the mass issue function processes all of the cardholder accounts that have been added as new issues.

Each cardholder account can be configured to produce any combination of a plastic card, a PIN mailer, and a card carrier (only one of these three, any two of these three, or all three). The items generated for each cardholder account depend upon the configuration in the CPF and CAF records. If the PIN MAILER CONTROL and CARD CARRIER CONTROL fields on screen 20 of the CPF are set to allow generation of PIN mailers and card carriers, then these items are generated according to the respective values in the PIN MAILER REQUIRED and PLASTIC/CARD CARRIER OPTIONS fields on screen 20 of the CAF.

Example Two—Cardholder Forgets PIN

A card administrator needs to send a PIN mailer to a cardholder who has forgotten their PIN.

The card administrator logs on to the BASE24 CRT access system and proceeds to the Cardholder Authorization File (CAF). The card administrator reads the cardholder's account and proceeds to screen 20. On CAF screen 20, the card administrator updates the ISSUE STATUS field to a value of 02 (Force issue), the PIN MAILER REQUIRED field to a value of 1 (PIN mailer), and the PLASTIC/CARD CARRIER OPTIONS field to a value of 3 (No plastic card and no card carrier).

If the PIN mailer needs to be created as soon as possible and cannot wait for a mass issue to be performed by the end of the day or week (the mass issue card generation schedule is determined by the institution), a force issue can be executed to individually produce the PIN mailer. To do this, a system operator accesses the Card Management Functions (CMF) screen and performs an issue maintenance function to prepare the card for plastic card generation. The system operator then enters the information for the card to be force issued, and presses the **F2** key to initiate the force issue process. A PIN mailer is produced without generating a plastic card or a card carrier.

Example Three—Lost or Stolen Card

A card administrator needs to flag a card account maintained in the BASE24 database as lost or stolen and generate a new card and a new PIN.

The card administrator logs on to BASE24 CRT access system and proceeds to the Cardholder

Authorization File (CAF). The card administrator reads the card account to reissue and updates the CARD STATUS field on CAF screen 1 to a value of 2 (Lost) or 3 (Stolen). A new CAF record is then added for the cardholder (either with the same card number but a new member number or with a new card number) with all of the cardholder's information from the lost/stolen CAF record. The ISSUE STATUS field on CAF screen 20 for this new CAF record is set to a value of 02 (Force issue).

If the plastic card needs to be created as soon as possible and cannot wait for a mass issue to be performed by the end of the day or week (the mass issue card generation schedule is determined by the institution), a force issue can be executed to individually produce the card. To do this, a system operator proceeds to the Card Management Functions (CMF) screen and performs an issue maintenance function to prepare the card for generation. The system operator then enters the information for the card to be force issued, and presses the **F2** key to initiate the force issue process. In this example, the card administrator would probably configure the cardholder account record so that a card and a card carrier would be produced. If the cardholder is also receiving a new PIN, the card administrator could select the PIN mailer option as well.

Example Four—Automatic Reissue

A group of cards is set to expire in 60 days and the institution needs to automatically set up the new card production.

The card management system administrator at the central office has been running the CMS Reissue Projection Report and from it knows that one thousand cards are set to expire in two months for a particular prefix.

In order that new cards are produced and mailed in time, the system administrator has set the production/reissue cycle fields in the Card Prefix File (CPF) to 60 days. During the daily or weekly schedule of card production, when the issue maintenance function is performed for the institution, the cards will be in their production/reissue cycle and be evaluated as to whether the cards should be reissued. If the cards pass the reissue criteria defined in the CPF and CAF, the appropriate cardholder accounts are updated with an issue status of 07 (Reissue). A subsequent run of the mass issue function produces the plastic cards for all of the cardholder accounts that have been flagged as reissues. In this example, the card administrator configures the cardholder account records so that plastic cards would be generated. If the cards will be mailed to the cardholders, the card carrier option would also be selected. If cardholders are keeping their existing PINs, the card administrator could choose not to produce PIN mailers.

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Section 3

Embossing, Encoding, and Indenting Formats

This section outlines the embossing, encoding, and indenting formats used in the BASE24-card product. The BASE24-card product supports card generation on the DataCard embossing machine.

Introduction

The BASE24-card product allows institutions the ability to configure the information embossed, encoded, and indented onto the plastic cards with a DataCard machine.

The embossing information is the data that is printed in raised characters on the card. The BASE24-card product supports up to four lines of emboss data.

The encoding information is the data that is held in the magnetic stripe on the back of the card. The BASE24-card product supports Track 1, Track 2 and Track 3 encoding.

The indenting information is the data that is indent printed in reverse-italic font on the back of the plastic card within the signature panel. The BASE24-card product supports one line of indent data.

Sentinels (special characters) are used to separate the information in the data stream that is sent to the DataCard embossing machine. The system formats the data stream as follows:

```
$emboss line 1 data *emboss line 2 data )emboss line 3 data  
#emboss line 4 data :indent data "%encode line 1 data ?;encode  
line 2 data ?encode line 3 data
```

The \$ sentinel indicates that emboss data is following. The : sentinel indicates that indent data is following. The “ sign before the sentinel for encode line 1 indicates that this is the start of the encoding section. Each sentinel tells the DataCard embossing machine what to do with the information following.

Card Embossing

Embossing control data can be defined on Card Prefix File (CPF) screen 21 for each card prefix defined in the system. The FORMAT TYPE field indicates how the data is formatted into the data stream. There are four different methods of formatting the emboss lines.

Note: Up to four lines of embossing information can be placed on a card. Each emboss line is limited to 50 bytes of data.

Format 01 (Template)

Format 01 (template) allows four lines of embossing information. A string of uppercase alphabetic characters determine the order, format, and content of various fields. The template strings contained in the LINE 1 FORMAT through LINE 4 FORMAT fields on CPF screen 21 determine the embossing data. The template is formatted and placed in the Embossing Control fields on CPF screen 21.

The following values are used with template formatting for embossing data, encoding magnetic track information, and formatting indenting data on the card:

“”	=	Text for up to 30 characters
A	=	Reserved for future use
B	=	Branch for 4 bytes
C	=	Issue date today for 6 bytes
D	=	Department for 2 bytes
E	=	Reserved for future use
F	=	Reserved for future use
G	=	Reserved for future use
H	=	Reserved for future use
I	=	Institution optional for 12 bytes
J	=	Primary account number for 20 bytes
K	=	Primary account number for 20 bytes
L	=	Primary account number for 20 bytes
M	=	Member number for 3 bytes
N	=	Card name for 26 bytes
O	=	PIN offset for 16 bytes
P	=	PAN for 19 bytes
Q	=	CVV for 3 bytes
R	=	CVV2 for 3 bytes
S	=	Card name 2 for 26 bytes
T	=	Primary account type for 2 bytes
U	=	SAN1 account type for 2 bytes
V	=	SAN2 account type for 2 bytes
W	=	Withdrawal limit for 4 bytes
X	=	Expiration date for 4 bytes

Y = Reserved for future use
 Z = Service code for 3 bytes

The syntax for having the data formatted using the template codes is to type a backslash (\), followed by a code, optionally followed by *offset.length*, and then followed by a space. For example, in order to emboss the card with the PAN, card name for 20 bytes, issue date for four bytes and expiration date the template would appear as follows:

`\P \N1.20 \C1.4 \X`

Which would translate to:

1234567890123456789JOHNATHAN PUBLIC JR10011202

It is not necessary to include the sentinels in the templates, since the BASE24-card program automatically formats these characters. Spacing of the data also needs to be formatted into the template. A total of 50 bytes of data can be placed in the data stream for each embossing line. This needs to be taken into consideration when building the template.

Format 02 (Default format + EXP + Work-Mbr + CVV Value + Service Code)

Format 02 includes a default format, expiration date, member number, CVV value, and service code. The default format consists of the following:

	1...5...0...5...0...5...0...5...0...5...0...5...0
Line #1	aaaaaaaaaaaaaaaaaaaaa
Line #2	bbbbbbbbbbbbbbbbbbbbbb

where:

a = PAN digits
 b = Cardholder name

Line three is embossed as:

Line #3	EXP cccc d eee fff
---------	--------------------

where:

c = Expiration date (z9/99)
 d = Member number
 e = CVV value (from CAF)
 f = Service code (from CAF)

Note: If the CAF service code is zeros, the literal 101 is used.

Example:

```
1234567890123456789
JOHNATHAN PUBLIC JR
EXP 12/02 0 123 456
```

Format 03 (Default Format + Exp Date)

Format 03 includes the default format and expiration date. The default format consists of the following:

```

1...5....0....5....0....5....0....5....0....5....0
Line #1 |aaaaaaaaaaaaaaaaaaaaa|
Line #2 |bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb|
```

where:

a = PAN digits

b = Cardholder name

Line three is embossed as:

```
Line #3 |cccc|
```

where:

c = Expiration date (z9/99)

Example:

```
1234567890123456789
JOHNATHAN PUBLIC JR
12/02
```

Format 00 (Default + Exp Date + Work-MBR)

Format 00 includes the default format, expiration date, and member number. The default format consists of the following:

```

1...5....0....5....0....5....0....5....0....5....0
Line #1 |aaaaaaaaaaaaaaaaaaaaa|
Line #2 |bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb|
```

where:

a = PAN digits

b = Cardholder name

Line three is embossed as:

```
Line #3 |cccc d
```

where:

c = Expiration date (z9/99)

d = Member number

Example:

1234567890123456789
JOHNATHAN PUBLIC JR
12/02 1

Card Encoding

Encoding control data can be defined on Card Prefix File (CPF) screen 21 and 22 for each card prefix defined in the system. The FORMAT TYPE field on screen 21 indicates how Track 1 and Track 2 data is formatted. There are two format type options for these tracks. The FORMAT CODE field on screen 22 dictates whether or not Track 3 is formatted. In order for Track 3 data to be formatted, template formatting of Track 1 and Track 2 data must be performed. Default track formatting does not include Track 3.

Track 1 allows up to 76 bytes of data. Track 2 allows 38 bytes of data and Track 3 allows up to 105 bytes. The data stream is blank filled if the tracks are not long enough. Through the use of the sentinel characters, the DataCard machine knows how to write the tracks.

Track 1 and Track 2

The following paragraphs describe the card encoding formats available for Track 1 and Track 2 data.

Format 01 (Template). Format 1 enables the customer to set up templates for Track 1 and Track 2 in the same manner as embossing templates. Card encoding templates use the same codes as defined for embossing templates using Format 01. The track templates are placed in the encoding control fields on CPF screen 21.

Track 1 template information is preceded by a the value B and concluded by the end sentinel. This information is placed in the TRACK1 FORMAT field on CPF screen 21.

Track 2 template information is concluded by the end sentinel. No preceding characters are needed. This information is placed in the TRACK2 DISCRETIONARY DATA FORMAT field on CPF screen 21.

Sentinel characters do not need to be included in the template. They are added programmatically.

Format 00 (Default Formatting). Format 2 provides default formats for Track 1 and Track 2. The default format for Track 1 consists of the following:

```
1...5....0....5....0....5....0....5....0....5....0....5....0
|aaaaaaaaaaaaaaaaaaaaa^bbbbbbbbbbbbbbbbbbbbbbbbbbbbbb^ccccddd
```

where:

- a = PAN digits (variable length)
- b = Cardholder name
- c = Expiration date
- d = Service code (from CAF)

Note: If the CAF service code is zeros, the literal 101 is used.

Example:

a = 1234567890123456789
 b = JOHN Q. PUBLIC
 c = 0212
 d = (CAF service code is zeros)

The Track 1 layout including sentinels and fields separators would be:

%B1234567890123456789^JOHN Q. PUBLIC^0212101?

The default format for Track 2 consists of the following:

1...5....0....5....0....5....0....5....0....5....0....5....0
 |aaaaaaaaaaaaaaaaaaaaa=bbbbccc

where:

a = PAN digits (variable length)
 b = Expiration date
 c = Service code (from CAF)

Note: If the CAF service code is zeros, the literal 101 is used.

Example:

a = 1234567890123456789
 b = 0212
 c = (CAF service code is zeros)

The Track 2 layout including sentinels and fields separators would be:

;1234567890123456789=0212101?

Track 3

Track 3 data can be formatted only if template formatting is used for Track 1 and Track 2 data. After formatting the first two tracks, the FORMAT CODE field on CPF screen 22 is evaluated to determine whether Track 3 data should be generated. If so, the Track 3 data is formatted in accordance with the ISO 4909 standard, ***Magnetic Stripe Data Content for Track 3***. A summary list of the information included in Track 3 data is shown below:

Start sentinel (_;)
 Format code (01)
 PAN
 Field separator
 Country code

Currency code
Currency exponent
Amount authorized per cycle period
Amount remaining this cycle
Cycle begin date
Cycle length
Retry count
PIN control parameters (algorithm ID, verification value)
Interchange control
Primary account type
Type of account and service restrictions SAN-1
Type of account and service restrictions SAN-2
Expiration date
Card sequence number (Member number)
Field separator
First subsidiary account number (SAN-1)
Field separator
Second subsidiary account number (SAN-2)
Field separator
Relay marker
Crypto check digits (Atalla offset)
End sentinel (?)

Card Indenting

Indent printing control data can be defined on Card Prefix File (CPF) screen 22 for each card prefix defined in the system. The FORMAT TYPE field indicates how the data is formatted into the data stream. There are three different methods of formatting the indent line, each of which is described below.

Note: The indent line is limited to 50 bytes of data.

Format 00 (No Indent Printing)

Format 00 enables the customer to turn off the indenting option. With this format type, no indenting data is formatted and placed on the card.

Format 01 (Template)

Format 01 enables the customer to set up templates for indenting in the same manner as embossing templates. Card indenting templates use the same codes as defined for embossing templates using format 01. The indent template is placed in the INDENT FORMAT field on CPF screen 22.

The indent template information is preceded by a colon (:) sentinel, although this sentinel character does not need to be included in the template because it is added programmatically. A total of 50 bytes of data can be placed in the data stream for the indenting line. This needs to be taken into consideration when building the template. The values are indented on the back of the plastic card within the signature panel in reverse-italic font.

Format 02 (Default Format—CVV2 only)

Format 02 enables the customer to indent print only the CVV2 value. No other values are formatted within the indenting data. The CVV2 value is indented on the back of the plastic card within the signature panel in reverse-italic font.

Bank Stock Header

If output of the data stream is to a tape or disk file, the Bank Stock Header is written at the beginning of each file. The Bank Stock Header consists of the vendor, FIID, stock, and prefix associated with the card prefix being run. Each time the vendor changes between prefixes for either tape or disk output, a new file is created and a new Bank Stock Header is written. If a stock changes between prefixes for tape output, a new Bank Stock Header is written, but the same file is used. If a stock changes between prefixes for disk output, a new Bank Stock Header is written and a new file is created.

Emboss Header

After the Bank Stock Header is written to the output file, an Emboss Header is written to the output file. This information explains the layout of the card information and dictates where the emboss and indent lines are to be placed on the actual card. The Emboss Header is hard coded for the BASE24-card product as shown and described below, where *b* is a blank space.

```
#DCL#080400b1#FC2#0580b2#FC1#0400b3#FC1#0270b4#FC1#0175b5#FC4#1250#END#@@@@@b0083
```

#DCL	Stands for Device Control Line Data.
#08	Indicates the card size.
0400	Indicates that the left margin on the card is 0.400 inches.
b1	Indicates the beginning of the first line of the line location record.
#FC2	Instructs the machine to print in font 2 CPI. Valid font values are 2–7).
#0580	Indicates that this line should be placed at 0.580 inches from the bottom of the card.
b2	Indicates the beginning of the second line of the line location record.
#FC1	Instructs the machine to print in font 1 CPI. Valid font values are 1–10.
#0400	Indicates that this line should be placed at 0.400 inches from the bottom of the card.
b3	Indicates the beginning of the third line of the line location record.
#FC1	Instructs the machine to print in font 1 CPI. Valid font values are 1–10.
#0270	Indicates that this line should be placed at 0.270 inches from the bottom of the card.

<i>b</i> 4	Indicates the beginning of the fourth line of the line location record.
#FC1	Instructs the machine to print in font 1 CPI. Valid font values are 1–10.
#0175	Indicates that this line should be placed at 0.175 inches from the bottom of the card.
<i>b</i> 5	Indicates the beginning of the indent print font and location information.
#FC4	Instructs the machine to indent print in font 4 CPI. Valid font values are 1–10.
#1250	Indicates that this line should be placed at 1.250 inches from the bottom of the card.
#END#@#@#@#@	Specifies the end of the Device Control Line Data.
0083	Indicates the length of the emboss header (the length can be up to 999999).

Data Layout

Following is the data layout of the emboss, encode, and indent information, the emboss header information, and the bank stock header information.

```

01 DET-REC.
02 EMBOSS-FORMAT.
03 LINE-1.
04 STR-CHAR          PIC X      VALUE "$".
04 EMB-LIN-1         PIC X(50).
03 LINE-2.
04 STR-CHAR          PIC X      VALUE "*".
04 EMB-LIN-2         PIC X(50).
03 LINE-3.
04 STR-CHAR          PIC X      VALUE ")".
04 EMB-LIN-3         PIC X(50).
03 LINE-4.
04 STR-CHAR          PIC X      VALUE "#".
04 EMB-LIN-4         PIC X(50).
02 INDENT-FORMAT.
03 STR-CHAR          PIC X      VALUE ":".
03 IND-LIN           PIC X(50).
02 ENCODE-STR-CHAR   PIC X      VALUE " ".
02 MAG-STRIPE.
03 IATA-STRIPE.
04 STR-SENTINAL      PIC X      VALUE "%".
04 TRACK-1-DATA      PIC X(76).
03 FILLER            PIC X      VALUE " ".
03 ABA-STRIPE.
04 STR-SENTINAL      PIC X      VALUE ";".
04 TRACK-2-DATA      PIC X(38).
03 FILLER            PIC X      VALUE " ".
03 TRK3-STRIPE.
04 TRACK-3-DATA      PIC X(105).
03 FILLER            PIC X      VALUE " ".
02 FILLER            PIC X(5)   VALUE SPACES.
01 WS-LINES-TO-EMBOSS PIC 9(01) VALUE ZEROES.

```

```

01 EMOSS-HEADER-1.
* 05 SEQ-NUM          PIC 9(06) VALUE 010000.
  05 HEADER          PIC X(11)
    VALUE "#DCL#080400".
*      | | | -LEFT MARGIN (0.400 IN)
*      | | | ---CARD SIZE
*      |-----DEVICE CONTROL:LINE DATA
  05 LINE-0          PIC X(11)
    VALUE " 1#FC2#0580".
*      | | | -TO BOTTOM OF CARD (0.580 IN)
*      | | | -----FONT 2 - 7 CPI
*      |-----LINE NUMBER
  05 LINE-1          PIC X(11)
    VALUE " 2#FC1#0400".
*      | | | -TO BOTTOM OF CARD (0.450 IN)
*      | | | -----FONT 1 - 10 CPI
*      |-----LINE NUMBER
  05 LINE-2          PIC X(11)
    VALUE " 3#FC1#0270".
*      | | | -TO BOTTOM OF CARD (0.350 IN)
*      | | | -----FONT 1 - 10 CPI
*      |-----LINE NUMBER
  05 LINE-3          PIC X(11)
    VALUE " 4#FC1#0175".
*      | | | -TO BOTTOM OF CARD (0.175 IN)
*      | | | -----FONT 1 - 10 CPI
*      |-----LINE NUMBER
  05 INDENT          PIC X(11)
    VALUE " 5#FC4#1250".
*      | | | -TO BOTTOM OF CARD (1.250 IN)
*      | | | -----FONT OCR Indent, 14 CPI
*      |-----LINE NUMBER
  05 TRAILER         PIC X(11)
    VALUE "#END#000000".

```

```

01 EMOSS-HEADER-1-LEN PIC S9(04) VALUE 83.

```

```

01 BANK-STOCK-HDR.
  05 GEN-KEY.
    10 VENDOR          PIC X(1).
    10 FIID            PIC X(4).
    10 STOCK           PIC X(1).
    10 PREFIX          PIC 9(11).
  05 FILLER            PIC X(63) VALUE SPACES.

```

Overall Example

The following is an example of all the embossing, encoding, and indenting information the DataCard machine needs to emboss and encode a plastic card.

Bank Stock Header:

ABNK2A0000000010000

Emboss Header:

#DCL#080400 1#FC2#0580 2#FC1#0400 3#FC1#0270 4#FC1#0175 5#FC4#1250#END#@000000 0083

Detail Data Record:

\$12345999999	JOHN Q.*12345999999	JOHN Q.)123459999
99	JOHN Q. PUBLIC#12345999999	JOHN Q.:12345999999
	JOHN Q."%B123459999	

This will emboss line one as:	12345999999	JOHN Q.
This will emboss line two as:	12345999999	JOHN Q.
This will emboss line three as:	12345999999	JOHN Q. PUBLIC
This will emboss line four as:	12345999999	JOHN Q.
This will indent print line as:	12345999999	JOHN Q.
This will encode Track 1 as:	123459999	

Section 4

Card Reporting

The BASE24-card product provides all reports and output required to manage a plastic card base for an institution. Customer information for the BASE24-card system is stored and updated on a Tandem computer. Card production control and issue reports are generated by the system. The remainder of this section describes the report generation process and each report format.

Card Reports

Reports are generated to aid in system control and can be divided into two categories: production control reports and issue reports. Both of these categories are described below.

Production Control Reports

Production control reports are produced by the mass issue and force issue functions of the Plastic Card Generation Server (SVPCGEN) process. They allow the financial institution to retain a hard copy record of the cards that were sent to be created, exceptions encountered during the run, and mailers to be sent to customers. Production control reports include the following:

- Card Production Control Report
- PCGEN Exception Report
- Card Carrier Report
- PIN Mailer Report

Issue Reports

Issue reports provide information pertinent to the issue and reissue of plastic cards. There are four reports; each produced by a different function. The issue reports and their means of generation are as follows:

- Available Account Numbers Report/Labels (Account Number Generation Server (SVAGEN))
- CMS Reissue Projection Report (Issue Projection function of the Card Management Function Server (SVCMPF) process)
- CMS Issue Maintenance Report (Issue Maintenance function of the Card Management Function Server (SVCMPF) process)
- CMS Reissue Exception Report (Issue Maintenance function of the Card Management Function Server (SVCMPF) process)

Generating the Reports

As described above, BASE24-card reports are generated by different functions. The Available Account Number Report and CMS Reissue Projection Report are a direct result of executing a report generating function. The other BASE24-card reports are generated to provide information about a plastic card generation function.

The following table shows the functions that cause the generation of each report. The first column of the table lists each report. The second column indicates the screen from which the function is initiated, and the third column lists the functions that generate the report, including the function key pressed to initiate the function.

Report	CRT Terminal	Function Key
Available Account Numbers Report	COF	F2 - Account Number Generation
Card Production Control Report	CMF	F2 - Force Issue F3 - Mass Issue
Card Carrier Report	CMF	F2 - Force Issue F3 - Mass Issue
CMS Issue Maintenance Report	CMF	F5 - Issue Maintenance
CMS Reissue Exception Report	CMF	F5 - Issue Maintenance
CMS Reissue Projection Report	CMF	F4 - Issue Projection
PCGEN Exception Report	CMF	F2 - Force Issue F3 - Mass Issue
PIN Mailer Report	CMF	F2 - Force Issue F3 - Mass Issue

Output Locations

BASE24-card reports can be generated to a disk file or Tandem spooler location. The output location is defined by an LCONF assign for each report. The following table lists the LCONF assigns associated with each report:

<u>LCONF Assign</u>	<u>BASE24-card Report</u>
ACCOUNT-NUM-RPT	Available Account Numbers Report
CARD-PRODUCTION-RPT	Card Production Control Report
CARD-CARRIER-RPT	Card Carrier Report
ISSUE-MAINT-RPT	CMS Issue Maintenance Report
EXCEPTION-RPT	CMS Reissue Exception Report
ISSUE-PROJ-RPT	CMS Reissue Projection Report
PCGEN-EXCEPTION-RPT	PCGEN Exception Report
PIN-MAILER-RPT	PIN Mailer Report

Available Account Numbers Report/Labels

The BASE24-card product aids the card-issuing institution by producing the Available Account Numbers Report. This report lists the possible account numbers institution personnel can assign to new cardholders. Possible available account numbers are listed for a specific prefix selected by the operator when performing the account number generation function.

The Account Number Generation Server (SVAGEN) is initiated online from the Card Operator Functions (COF) screen, part of the BASE24-card online entry system. When generating available account numbers, the operator can specify whether the account numbers are listed on a report or on labels.

A sample of the Available Account Numbers Report is shown below, followed by descriptions of the fields that appear on the report.

AVAILABLE ACCOUNT NUMBERS REPORT					
INSTITUTION ID NUMBER			DATE DD MMM YY		
BNK2			PAGE 01		
CARD ACCOUNT NUMBER	CARDHOLDER NAME	BANK EMPLOYEE NAME	ACCOUNT DATE (DD/MM/YY)	OPENING TIME (HH:MM)	
1111100000000150	-001		/	/	:
1111100000000165	-001		/	/	:
1111100000000173	-001		/	/	:
1111100000000187	-001		/	/	:
1111100000000197	-001		/	/	:
1111100000000204	-001		/	/	:
1111100000000210	-001		/	/	:
1111100000000227	-001		/	/	:
1111100000000231	-001		/	/	:
1111100000000243	-001		/	/	:
1111100000000258	-001		/	/	:
1111100000000266	-001		/	/	:
1111100000000279	-001		/	/	:
***** END OF REPORT *****					

INSTITUTION ID NUMBER — The FIID of the institution (e.g., BNK2) for which available account numbers were generated.

CARD ACCOUNT NUMBER — The possible account numbers for assignment.

CARDHOLDER NAME — This blank column is provided to fill in the name of the individual receiving the corresponding account number at the time an account number is assigned.

BANK EMPLOYEE NAME — This blank column is provided to fill in the name of the institution employee assigning the corresponding account number at the time an account number is assigned.

ACCOUNT OPENING DATE — This blank column is provided to fill in the date that the corresponding account number was assigned.

ACCOUNT OPENING TIME — This column is provided to fill in the time that the corresponding account number was assigned.

Card Production Control Report

The mass issue and force issue functions are initiated from the Card Management Functions (CMF) screen, the Plastic Card Generation Server (SVPCGEN) process produces a file containing the information required to emboss, encode, and indent every plastic card identified for issue in the run. The SVPCGEN process also generates the Card Production Control Report, which provides information about the cards that will be produced.

Detail records in this report summarize items (cards, PIN mailers, or card carriers) that were successfully produced. If multiple items were requested, but only some of them were produced, then a *partially successful* completion situation exists. An asterisk (*) to the left of a detail record in this report indicates a partially successful completion. For each partially successful completion, corresponding detail records are printed on the PCGEN Exception Report, which explain any errors encountered. Records on the PCGEN Exception Report that are identified with an asterisk (*) explain which items were requested and created for this cardholder account record. In summary, cards for which all requested items were successfully completed are listed on the Card Production Control Report only. Cards for which processing failed are listed on the PCGEN Exception Report only. Cards for which only some items were successfully completed are listed on both reports.

A sample of the Card Production Control Report is shown below, followed by descriptions of the fields that appear on the report.

CARD PRODUCTION CONTROL REPORT									
INSTITUTION ID BNK2		TEST BANK 2		DATE DD MMM YY PAGE 1					
EMBOSSED CARD NUMBER	EMBOSSED MEM NUMBER	P M	C C	CARDS ISSUED	EMBOSSED CARDHOLDER NAME	R E C E I V E D B Y :		DATE	
1111111111000007	-000	N	Y	0	JONATHAN SMITH	-----			
1111111111000008	-000	Y	N	0	SAMATHA JONES	-----			
1111111111000009	-000	N	N	1	CHARLIE SAMSON	-----			
1111111111000352	-000	Y	Y	0	SUSIE SWENSON	-----			
1111111111098009	-000	Y	N	1	JULIE HAYES	-----			
* 1111111111009800	-000	N	Y	1	RAY WHITE	-----			
1111111111000089	-000	Y	Y	1	MIKE THOR	-----			
****	TOTAL NUMBER OF PLASTICS ONLY				1	****			
****	TOTAL NUMBER OF PIN MAILERS ONLY				1	****			
****	TOTAL NUMBER OF CARD CARRIERS ONLY				1	****			
****	TOTAL NUMBER OF PIN MAILERS AND CARD CARRIERS				1	****			
****	TOTAL NUMBER OF PIN MAILERS AND PLASTICS				1	****			
****	TOTAL NUMBER OF CARD CARRIERS AND PLASTICS				1	****			
****	TOTAL NUMBER OF PLASTICS, CARD CARRIERS, PIN MAILERS				1	****			
***** END OF REPORT *****									

INSTITUTION ID — The FIID of the institution for which the mass issue or force issue was requested. The institution name is printed to the right of the FIID.

EMBOSSED CARD NUMBER — The cardholder number to be embossed on the plastic card.

EMBOSSSED MEM NUMBER — The cardholder member number to be embossed on the plastic card.

PM — A flag indicating whether PIN mailer data was created for this cardholder when the plastic card data was created. Values are as follows:

Y = Yes, PIN mailer data was created for this cardholder.

N = No, PIN mailer data was not created for this cardholder.

CC — A flag indicating whether card carrier data was created for this cardholder when the plastic card data was created. Values are as follows:

Y = Yes, card carrier data was created for this cardholder.

N = No, card carrier data was not created for this cardholder.

CARDS ISSUED — The number of plastic cards issued for the cardholder in this run.

EMBOSSSED CARDHOLDER NAME — The cardholder name to be embossed on the plastic card.

RECEIVED BY — This blank column is provided to fill in the name of the person who receives the plastic card.

DATE — This blank column is provided to fill in the date that the cardholder receives the plastic card.

TOTAL NUMBER OF PLASTICS ONLY — The total number of cardholder accounts for which only card data (no card carrier or PIN mailer data) was produced during this mass issue or force issue run.

TOTAL NUMBER OF PIN MAILERS ONLY — The total number of cardholder accounts for which only PIN mailer data (no card or card carrier data) was produced during this mass issue or force issue run.

TOTAL NUMBER OF CARD CARRIERS ONLY — The total number of cardholder accounts for which only card carrier data (no card or PIN mailer data) was produced during this mass issue or force issue run.

TOTAL NUMBER OF PIN MAILERS AND CARD CARRIERS — The total number of cardholder accounts for which PIN mailer data and card carrier data (no card data) was produced during this mass issue or force issue run.

TOTAL NUMBER OF PIN MAILERS AND PLASTICS — The total number of cardholder accounts for which PIN mailer data and card data (no card carrier data) was produced during this mass issue or force issue run.

TOTAL NUMBER OF CARD CARRIERS AND PLASTICS — The total number of cardholder accounts for which card carrier data and card data (no PIN mailer data) was produced during this mass issue or force issue run.

TOTAL NUMBER OF PLASTICS, CARD CARRIERS, PIN MAILERS — The total number of cardholder accounts for which card data, card carrier data, and PIN mailer data was produced during this mass issue or force issue run.

Card Carrier Report

A card carrier is a specially designed form used to mail plastic cards to cardholders. The carrier has a pocket or envelope for insertion of the newly created card. Card carriers commonly contain preprinted information such as the institution's name, address, and logo, as well as instructions for the customer concerning lost or stolen cards.

The Card Carrier Report is generated during execution of a mass issue or force issue function initiated from the Card Management Function (CMF) screen. The BASE24-card product allows each institution to choose whether or not card carriers are printed during the card generation process. A card carrier is written to the Card Carrier Report for the card being issued if the PLASTIC/CARD CARRIER OPTIONS field on CAF screen 20 is set to a value of 1 or 2 for the cardholder account and the CARD CARRIER CONTROL field on CPF screen 20 is set to a value of 1 for the corresponding CPF record.

A sample Card Carrier Report is shown below, followed by descriptions of the fields that appear on the report.

NAME1 ADDR1 ADDR2 (if not spaces) CITY ST CODE COUNTRY (see note)	YOUR CARD NUMBER CARD NUM CARD LIMIT \$****0.00 EXPIRATION DATE MMYYY CARDS ISSUED ##
---	---

CARD NUM — The primary account number (PAN) from the CAF.

CARD LIMIT — The card limit (\$****0.00) for this card. This value is taken from the TOTAL AGGR field on CAF screen 1 for this cardholder account.

EXPIRATION DATE — The expiration date (MMYY) of the card. This value corresponds to the value from the EXPIRATION DATE field on CAF screen 2 for the cardholder account.

CARDS ISSUED — The total number of cards (##) to be issued for this cardholder account. This value represents the value from the CARDS TO ISSUE field on CAF screen 20 for this cardholder account.

NAME1 — The name of the primary cardholder. This is the value from the NAME LINE1 field on CAF screen 20 for this cardholder account.

ADDR1 — The first line of the cardholder's address. This is the value from the ADDR LINE1 field on CAF screen 20 for this cardholder account.

ADDR2 — The second line of the cardholders' address. This is the value from the ADDR LINE2 field on CAF screen 20 for this cardholder account.

CITY — The city of residence for the cardholder. This is the value from the CITY field on CAF screen 20 for this cardholder account.

ST-CODE — The state and ZIP code of residence for the cardholder. These values are from the STATE and CODE fields on CAF screen 20 for this cardholder account.

COUNTRY — The country of residence for the cardholder. This value is from the COUNTRY field on CAF screen 20 for this cardholder account.

Note: The country is printed on the Card Carrier Report only if the country code in the CAF record is different than the country code for the FIID associated with that CAF record. (The country code for the FIID is defined in the IDF.)

CMS Issue Maintenance Report

Prior to actual issue of plastic cards, the issue maintenance function initiated from the Card Management Function (CMF) screen reviews the status of all card accounts. During this review, each account is checked to determine if a new plastic card is to be created in the current run.

Accounts are selected for card issuance for a number of reasons. For example, cards are produced for new accounts, for accounts with cards that have expired, to replace lost or stolen cards, or for an institution's entire card base. Once an account is selected for card production, the CMF server ensures that all required data for producing the plastic card exists in the BASE24 files in a valid format. Accounts that pass the issue requirements, as defined by the institution, are listed on the CMS Issue Maintenance Report.

A sample of the CMS Issue Maintenance Report is shown below, followed by descriptions of the fields that appear on the report.

CMS ISSUE MAINTENANCE REPORT										
INSTITUTION ID /NAME								DATE DD MMM YYYY		
ID NUMBER BNK2	TEST BANK 2							01		
CARD NUMBER	MEM	CARDHOLDER NAME	CARD TYPE	LAST MAINTENANCE DATE	DATE	TIME	EXP	DATE OF FIRST USE	DATE OF LAST USE	ISSUE STATUS
1111111100000007	-000	JONATHAN SMITH	PROP	17 SEP 01	08:48	JAN 03	12 JAN 00	31 AUG 99	REISSUE	NOT ACT
1111111100000008	-000	SAMATHA JONES	PROP	16 SEP 01	12:34	DEC 02		31 DEC 74	NEW ISSUE	OPEN
1111111100000009	-000	CHARLIE SAMSON	PROP	16 SEP 01	13:17	JAN 02	14 JUN 00	01 SEP 01	REISSUE	OPEN
***** END OF REPORT *****										

ID NUMBER — The FIID of the institution for which the issue maintenance function was requested. The institution name is printed to the right of the FIID.

CARD NUMBER — The plastic card number passing issue requirements.

MEM — The member number of the plastic card.

CARDHOLDER NAME — The name to appear on the front of the plastic card.

CARD TYPE — A code identifying the type of card associated with the card number to be produced. This value corresponds to the value from the CARD TYPE field on CAF screen 1 for the cardholder account.

LAST MAINTENANCE DATE — The date (DD MMM YY) of the last file maintenance operation performed on the corresponding CAF record.

LAST MAINTENANCE TIME — The time (hh:mm) of the last file maintenance operation performed on the corresponding CAF record.

DATE EXP — The expiration date (MMM YY) of the card. This value corresponds to the value from the EXPIRATION DATE field on CAF screen 2 for the cardholder account.

DATE OF FIRST USE — The date (DD MMM YY) the corresponding CAF record was first used in the BASE24 system.

DATE OF LAST USE — The date (DD MMM YY) the corresponding CAF record was last used in the BASE24 system.

ISSUE STATUS — The issue status of the card being produced. This value corresponds to the value from the ISSUE STATUS field on CAF screen 20 for the cardholder account.

CARD STATUS — The card status of the card being produced. This value corresponds to the value from the CARD STATUS field on CAF screen 1 for the cardholder account.

CMS Reissue Exception Report

The issue maintenance function initiated from the Card Management Function (CMF) screen reviews the status of all card accounts. During this review, each account is checked to determine if a new plastic card is to be created in the current run. The issue maintenance function is required to be performed before the actual creation of the plastic cards.

Accounts are selected for card issuance for a number of reasons. For example, cards are produced for new accounts, for accounts with cards that have expired, to replace lost or stolen cards, or for an institution's entire card base. Once an account is selected for card production, the CMF server ensures that all required data for producing the plastic card exists in the BASE24 files in a valid format. Accounts not passing the issue requirements, as defined by the institution, are listed on the CMS Reissue Exception Report.

A sample of the CMS Reissue Exception Report is shown below, followed by descriptions of the fields that appear on the report.

CMS REISSUE EXCEPTION REPORT										
INSTITUTION ID NUMBER BNK2			DATE DD MMM YYYY							
TEST BANK 2			01							
CARD NUMBER	MEM	CARDHOLDER NAME	CARD TYPE	LAST MAINTENANCE DATE	DATE TIME EXP	DATE OF FIRST USE	DATE OF LAST USE	ISSUE STATUS	CARD STATUS	
1111111100000001	-000	HENRY COOK	PROP	17 SEP 01 08:48	JAN 04 12	JAN 00	31 AUG 01	NEW ISSUE	CLOSED	
2222222200000012	**	CAF CARD STATUS NOT INCLUDED ON CPF CRD STAT	PROP	16 SEP 01 12:34	DEC 00		31 DEC 74	EXPIRED	OPEN	
3434343400000039	-000	CHARLIE SAMSON	PROP	16 SEP 01 13:17	JUL 01 14	JUN 00	01 SEP 01	PURGED	OPEN	
	**	CAF RECORD IS OUT OF EXPIRATION CYCLE	**	-- THIS RECORD HAS BEEN PURGED FROM CAF						
***** END OF REPORT *****										

INSTITUTION ID NUMBER — The FIID of the institution for which the issue maintenance function was requested. The institution name is printed to the right of the FIID.

CARD NUMBER — The plastic card number failing issue requirements.

MEM — The member number of the plastic card.

CARDHOLDER NAME — The name to appear on the front of the plastic card.

CARD TYPE — A code identifying the type of card associated with the card number to be produced. This value corresponds to the value from the CARD TYPE field on CAF screen 1 for the cardholder account.

LAST MAINTENANCE DATE — The date (DD MMM YY) of the last file maintenance operation performed on the corresponding CAF record.

LAST MAINTENANCE TIME — The time (hh:mm) of the last file maintenance operation performed on the corresponding CAF record.

DATE EXP — The expiration date (MMM YY) of the card. This value corresponds to the value from the EXPIRATION DATE field on CAF screen 2 for the cardholder account.

DATE OF FIRST USE — The date (DD MMM YY) the corresponding CAF record was first used in the BASE24 system. This is the date from the DATE FIRST USED field on CAF screen 2 for the cardholder account.

DATE OF LAST USE — The date (DD MMM YY) the corresponding CAF record was last used in the BASE24 system.

ISSUE STATUS — The issue status of the card being produced. This value corresponds to the value from the ISSUE STATUS field on CAF screen 20 for the cardholder account.

CARD STATUS — The card status of the card being produced. This value corresponds to the CARD STATUS field on CAF Screen 1.

EXCEPTION and REMEDY — This line explains why the above record was an exception and lists a possible remedy.

CMS Reissue Projection Report

The CMS Reissue Projection Report is generated when the issue projection function is initiated from Card Management Functions (CMF) screen for a specified institution. This report lists the number of plastic cards due for reissue each month for an individual institution. It lists each card type separately, and covers a 12-month range, beginning with the current month of the current year and ending with previous month of the following year. For example, if you are running this report in September of 2001, it starts with SEP 2001 and ends with AUG 2002.

The CMS Reissue Projection Report can be used for estimating the quantity of materials needed for output production.

The report lists valid card types in the first column. The remaining columns list the total number of cards due for reissue for each month in the range for the card type. The last column on the report lists the aggregate total of cards to be reissued for the card type for the entire year.

A sample of the CMS Reissue Projection Report is shown below.

CMS REISSUE PROJECTION REPORT													
INSTITUTION ID NUMBER BNK2	DATE DD MM YYYY												01
TEST BANK 2													
CARD TYPE	SEP 2001	OCT 2001	NOV 2001	DEC 2001	JAN 2002	FEB 2002	MAR 2002	APR 2002	MAY 2002	JUN 2002	JUL 2002	AUG 2002	12 MONTH TOTAL
P (PROP)	0	10	25	0	0	20	5	10	0	20	0	10	100
VV(VISA)	0	5	25	0	0	20	5	10	5	20	5	5	100
***** END OF REPORT *****													

PCGEN Exception Report

The mass issue and force issue functions initiated from the Card Management Functions (CMF) screen produce a disk or tape file containing the information required to emboss, encode, and indent every plastic card identified for issue in the run. A separate output file is also created for a PIN Mailer Report and Card Carrier Report, if a PIN mailer or card carrier was specified to be created for the card. If any card data items (cards, PIN mailers, or card carriers) cannot be successfully created, information is provided on the PCGEN Exception Report as to why the output file for the item could not be created.

Detail records in this report summarize processing problems. If a problem occurred while creating the data for a card, PIN, PIN mailer, or card carrier, a summary record (with an asterisk to the left) lists which items were requested and created.

If multiple items were requested, but only some of them were produced, then a *partially successful* completion situation exists. For each partially successful completion, a corresponding detail record appears on the Card Production Control Report, explaining which items were successfully produced.

In summary, cards for which all requested items were successfully completed are listed on the Card Production Control Report only. Cards for which all item processing failed are listed on the PCGEN Exception Report only. Cards for which only some items were successfully completed are listed on both reports.

A sample of the PCGEN Exception Report is shown below, followed by descriptions of the fields that appear on the report.

INSTITUTION ID		PCGEN EXCEPTION REPORT		DATE DD MMM YYYY	
BNK2	TEST BANK 2			PAGE 1	
GEN KEY	CARD NUMBER	MESSAGE			
ABNK2B000000011111	1111111100000017	MAIN - CAF ISSUE-STAT MUST BE 01 OR 02 OR 03 OR 07			
ABNK2B000000011111	111111111009800	MAILER - MAILER CREATION ERROR			
*ABNK2B000000011111	111111111009800	SUMMARY (REQUESTED/ACTUAL): MAILER Y/N, CARRIER Y/Y, PLASTIC 1 / 1			
ABNK2B00000022222		MAIN - NO RECORDS IN CAF FOR THE GEN-KEY			
*****		END OF REPORT *****			

INSTITUTION ID — The FIID of the institution for which the mass issue or force issue function was requested. The institution name is printed to the right of the FIID.

GEN KEY — The value of the GEN-KEY field from the CAF for which the exception occurred.

CARD NUMBER — The plastic card number for which the exception occurred (if applicable)

MESSAGE — The reason why the exception occurred.

PIN Mailer Report

PIN mailers prepared by the BASE24-card system inform cardholders of PINs to be used with the newly issued plastic cards. Using PIN mailers, customers are notified of PINs in a mailing separate from that of the plastic card. Should the card be lost or stolen before reaching the rightful cardholder, the PIN is not immediately available to the unauthorized cardholder.

A PIN mailer is a special form designed and ordered by the card-issuing institution. The institution can include preprinted information on the PIN mailer such as the institution's name, address and logo.

PIN mailers are produced when the mass issue or force issue functions are initiated from the Card Management Functions (CMF) screen and a PIN mailer needs to be produced for the cardholder. The BASE24-card product allows institutions to choose whether or not PIN mailers are printed using the card generating process. A PIN mailer is generated for the card being issued if the PIN MAILER REQUIRED field on CAF screen 20 is set to a value of 1 for the cardholder account record and the PIN MAILER CONTROL field on CPF screen 20 is set to a value of 1 in the corresponding card prefix record.

The PINs generated by the BASE24-card system and included in PIN mailers can be generated in software or in a hardware security module. If the LCONF param HRDWR-PIN-GEN is set to a value of N, PINs are generated in software. If this param is set to a value of Y, PINs are generated in a hardware security module. When PINs are generated in software, the PIN Mailer Report is written to a Tandem spooler location or a Tandem disk file. When PINs are generated in a hardware security module, PIN mailers are written to a printer directly connected to the hardware security module.

The format of PIN mailers vary slightly, depending on whether the PIN printed in the PIN mailer was generated in software or in hardware. Both formats are explained on the following pages.

Software PIN Mailer Report

A sample PIN Mailer Report for a PIN generated in software is shown below, followed by descriptions of the fields that appear on the report.

NAME1	
ADDR1	
ADDR2 (if not spaces)	
CITY	
ST-CODE	
COUNTRY (if not zeros)	
	PIN

NAME1 — The name of the cardholder. This corresponds to the value in the NAME LINE 1 field on CAF screen 20 for the cardholder account.

ADDR1 — The first line of the cardholder's address. This corresponds to the value in the ADDR LINE1 field on CAF screen 20 for the cardholder account.

ADDR2 — The second line of the cardholder's address, if applicable. This corresponds to the value in the ADDR LINE2 field on CAF screen 20 for the cardholder account.

CITY — The city of residence for the cardholder. This corresponds to the value in the CITY field on CAF screen 20 for the cardholder account.

ST-CODE — The state and ZIP code of residence for the cardholder. These correspond to the values in the STATE field and CODE field on CAF screen 20 for the cardholder account.

PIN — The new PIN generated by BASE24 software.

Hardware PIN Mailer

A sample PIN mailer for a PIN generated in hardware is shown below, followed by descriptions of the fields that appear on the mailer.

					PIN	
NAME1						
NAME2						
ADDR1						
ADDR2						
CITY	ST	ZIP	COUNTRY (see note)			

PIN — The new PIN generated by the hardware security module.

NAME1 — The name of the primary cardholder. This corresponds to the value in the NAME LINE1 field on CAF screen 20 for the cardholder account.

NAME2 — The name of the secondary cardholder. This corresponds to the value in the NAME LINE2 field on CAF screen 20 for the cardholder account.

ADDR1 — The first line of the cardholder's address. This corresponds to the value in the ADDR LINE1 field on CAF screen 20 for the cardholder account.

ADDR2 — The second line of the cardholder's address, if applicable. This corresponds to the value in the ADDR LINE2 field on CAF screen 20 for the cardholder account.

CITY — The city of residence for the cardholder. This corresponds to the value in the CITY field on CAF screen 20 for the cardholder account.

ST — The state of residence for the cardholder. This corresponds to the value in the STATE field on CAF screen 20 for the cardholder account.

ZIP — The postal code of or residence for the cardholder. This corresponds to the value in the POSTAL CODE field on CAF screen 20 for the cardholder account.

COUNTRY — The country of residence for the cardholder. This value is from the COUNTRY field on CAF screen 20 for this cardholder account.

Note: The country is printed on the PIN mailer only if the country code in the CAF record is different than the country code for the FIID associated with that CAF record. (The country code for the FIID is defined in the IDF.)